

An algebraic criterion for absolute irreducibility: completion

4. Norms of the auxiliary linear form

Let

$$p = ld^{n-1}, \quad q = md^{n-1},$$

so that these are the degrees of the one-variable polynomials $f(\xi)$ and $g(\xi)$ obtained above by Kronecker substitution. Write

$$\prod(\xi - a_i\eta) = \xi^p + r_1\xi^{p-1}\eta + \cdots + r_p\eta^p,$$

$$\prod(\xi - b_j\eta) = \xi^q + s_1\xi^{q-1}\eta + \cdots + s_q\eta^q,$$

and

$$\prod(\xi - c_k\eta) = \xi^{p+q} + t_1\xi^{p+q-1}\eta + \cdots + t_{p+q}\eta^{p+q}.$$

Thus r, s, t denote the homogenized elementary symmetric functions in the rows a and b ; equivalently, they are the symmetric functions, linear in each separate row, from which all integral symmetric functions that are homogeneous and of equal degree in each row are composed uniquely and integrally.

Introduce further indeterminates $u_{\mu\nu}$ and form the linear expression

$$\ell(u, a, b) = \sum u_{\mu\nu} a^\mu b^\nu, \tag{9}$$

where the summation is extended over prescribed index pairs (μ, ν) . Multiply $\ell(u, a, b)$ by all its conjugates obtained from permutations of the rows a, b , omitting repetitions. This product is the norm $N(\ell(u))$ of $\ell(u)$ with respect to the field of the $t(a, b)$, when $\ell(u)$ is regarded as a function in the field of the $r(a, b)$ and $s(a, b)$. It is an integral symmetric function of all rows a, b , homogeneous of the same degree in each row. Hence, by the preceding discussion, it is a uniquely determined integral polynomial in the $t(a, b)$ and the $u_{\mu\nu}$:

$$N(\ell(u, a, b)) = T(t(a, b), u), \tag{10}$$

identically in a, b, u .

The same argument applied to $z - \ell(u)$ gives

$$N(z - \ell(u)) = z^\delta + T_{\delta-1}(t, u)z^{\delta-1} + \cdots + T_0(t, u),$$

where all T_i are integral polynomials in t and u . Both sides vanish for $z = \ell(u)$. Because the elementary symmetric functions $r(a, b)$ and $s(a, b)$ are algebraically independent, the identity remains an identity in r, s, u once $t(a, b)$ is replaced by the corresponding integral bilinear expressions $w(r, s)$

and $\ell(u)$ is interpreted as a function of r and s . Thus one obtains

$$\ell(u)^\delta + T_{\delta-1}(w(r, s), u)\ell(u)^{\delta-1} + \cdots + T_0(w(r, s), u) = 0, \quad (11)$$

identically in r, s, u .

The identities (10) and (11) persist under every specialization of the systems a, b , and hence of r, s . Suppose in particular that the specialized systems are chosen so that all products $a^\mu b^\nu$ that occur in (9) vanish. Then $\ell(u)$ vanishes under the substitution. If $w(\rho, \sigma) = \tau$, equation (11) gives

$$T_0(\tau, u) = 0 \quad (12)$$

identically in u .

Conversely, let $\tau_1, \dots, \tau_{p+q}$ be a system, not all zero, for which (12) holds. In an algebraic extension field there is a factorization

$$\xi^{p+q} + \tau_1 \xi^{p+q-1} \eta + \cdots + \tau_{p+q} \eta^{p+q} = \prod_i (\xi - \alpha_i \eta) \prod_j (\xi - \beta_j \eta),$$

where in no factor do both α and β vanish simultaneously, although some individual α or β may vanish. Substitution in (10), using (12), shows that $\ell(u, \alpha, \beta)$ or one of its conjugate factors vanishes identically in u . After a suitable numbering of the α 's and β 's, putting $r(\alpha, \beta) = \rho$ and $s(\alpha, \beta) = \sigma$ means exactly that

$$\tau(\xi, \eta) = \rho(\xi, \eta)\sigma(\xi, \eta), \quad (13)$$

with all products that occur in $\ell(u)$ vanishing, but not all coefficients of ρ or of σ vanishing.

5. Construction of the reducibility equations

Now take p and q to be the degrees of $f(\xi)$ and $g(\xi)$ in the Kronecker substitution. Split the indices μ, ν into two classes. In the first class, μ runs through the exponents missing from $f(\xi)$ and ν through the exponents occurring in $g(\xi, \eta)$; in the second class, ν runs through the exponents missing from $g(\xi)$ and μ through the exponents occurring in $f(\xi, \eta)$.

Let

$$e(\xi) = \sum Z_k \xi^k$$

be the one-variable polynomial corresponding to the original polynomial $E(x)$ under Kronecker substitution. Define $S(Z, u)$ to be the integral polynomial in Z and u obtained from the right hand side $T(t, u)$ of (10), with summation restricted to the index classes just described, by replacing the t by the coefficients Z of $e(\xi)$ and by zeros in the positions of exponents absent from e .

If now one substitutes the coefficients A, B of F, G and the corresponding zeros of f and g for r, s , then the linear form $\ell(u)$ vanishes by the chosen summation. Moreover $t = w(r, s)$ becomes the coefficient system $C(A, B)$, together with the corresponding zeros of $h(\xi)$. Hence $T(t, u)$ becomes

$S(C(A, B), u)$. From (11) one obtains

$$S(C(A, B), u) = 0 \quad (14)$$

identically in A, B, u .

Since (14) remains true under specialization, the coefficient systems

$$\Gamma = C(A, B)$$

of all specialized forms $H(x)$, and more generally $H(x, y)$, that split into two factors in an extension field, satisfy

$$S(\Gamma, u) = 0 \quad (15)$$

identically in u .

Conversely, suppose that the coefficients Γ , not all zero, of a polynomial $H(x)$ or a form $H(x, y)$ satisfy (15). This is equivalent, by the preceding construction, to $T(\Gamma, u) = 0$ for the coefficients of the one-variable form $h(\xi, \eta)$. By (13) there is, in an algebraic extension field of the coefficient field, a factorization

$$h(\xi, \eta) = f(\xi, \eta)g(\xi, \eta),$$

where, by the restricted choice of indices, no exponents occur in f and g except those induced by Kronecker substitution. Therefore this one-variable factorization comes from a factorization of the original homogeneous form,

$$H(x, y) = F(x, y)G(x, y).$$

If no factor becomes of degree zero when $y = 1$, in particular if the coefficients Γ do not lower the degree of $H(x)$, one obtains also

$$H(x) = F(x)G(x).$$

Thus (15) is necessary and sufficient for $H(x, y)$, and, after exclusion of degree lowering, also $H(x)$, to split in an extension field.

It follows further that $S(Z, u)$ cannot vanish identically in Z and u . Section 1 showed that for $n > 2$ the polynomial with indeterminate coefficients, and therefore the corresponding homogeneous form in one further variable, is absolutely irreducible. Consequently

$$S(Z, u) = \sum_i \Phi_i(Z)U_i,$$

where the U_i are monomials in the u 's and, by (14), each $\Phi_i(Z)$ belongs to the prime ideal $\mathfrak{B}$ of all reducibility relations. We have therefore shown that $\mathfrak{B}$ has no zeros other than those represented by the parameterization $\Gamma = C(A, B)$ with A, B in an algebraic extension field. The converse asserted at the end of section 2 is proved.

6. The reducibility form

The condition for absolute irreducibility can now be formulated directly. Decompose the degree of $E(x)$ in every possible way as a sum of two positive integers $l + m$. For each such decomposition form the corresponding $S(Z, u)$. The product over all decompositions,

$$R(Z, u) = \prod S(Z, u), \quad (16)$$

is the reducibility form of $E(x)$. Indeed, by the preceding result every splitting of $E(x)$, respectively of $E(x, y)$, makes one factor of (16) vanish, and conversely. This proves the main theorem stated in the introduction, and at the same time shows how the reducibility form may actually be constructed in finitely many steps.

7. Ostrowski's theorem

The existence of the reducibility form (16) immediately gives the theorem of Ostrowski cited in the introduction. Let

$$E(x) = \sum \Gamma_i x^i$$

be an absolutely irreducible polynomial whose coefficients Γ_i are algebraic integers. Let K be a finite algebraic number field containing all the Γ_i , and let $\mathfrak{p}$ be a prime ideal of K .

For the exact degree of $E(x)$ the reducibility form $R(Z, u)$ does not vanish after the substitution $Z = \Gamma$; say

$$R(\Gamma, u) = \sum_j P_j U_j \neq 0$$

identically in u . The ideal $(P_1, P_2, \dots)$ is divisible only by finitely many prime ideals of K . Hence, except for these finitely many $\mathfrak{p}$, the reduction of $R(\Gamma, u)$ modulo $\mathfrak{p}$ is not identically zero. Since the residue classes modulo $\mathfrak{p}$ again form a field, it follows that $E(x)$ modulo $\mathfrak{p}$ is irreducible, indeed absolutely irreducible. The same conclusion holds for the homogenized form $E(x, y)$, so that no lowering of degree occurs modulo $\mathfrak{p}$.

If the coefficients of $E(x)$ are algebraic fractions rather than algebraic integers, one must also exclude the finitely many prime ideals that divide a common denominator. For every remaining prime ideal one may replace $E(x)$ by a scalar multiple with algebraic integral coefficients, and the preceding argument applies. This proves the theorem.

Formal calculus of variations and differential invariants

The generation of all differential invariants and the derivation of the principal theorems are most uniformly carried out by the formal methods of the calculus of variations. This principle goes back to Riemann and received its principal development, in form-theoretic terms, through Lipschitz.

Begin with the variational problem belonging to a homogeneous form $f(dx)$,

$$\delta \int f(dx) = 0. \quad (140)$$

By partial integration one is led to the invariant system of equations

$$\psi_i = 0,$$

where the Lagrange expressions ψ_i are contragredient to the differentials dx_i . Formally this is seen most simply by defining the ψ_i through the integration-free identity corresponding to partial integration, the Lagrange central equation,

$$df - dh = - \sum_i D\psi_i dx_i.$$

Here both f and df are invariants, and consequently the right hand side is invariant as well.

For a quadratic form one obtains in particular

$$\sum g_{ij} dx_i dx_j - 2 \sum g_{ij} dx_i D dx_j = -2 \sum_i \psi_i(d, d) D x_i.$$

If d is replaced by the cogredient expression $d + \lambda \delta$ and powers of λ are compared, one obtains the corresponding system of identities

$$\sum D g_{ij} dx_i dx_j - 2 \sum g_{ij} dx_i D dx_j = -2 \sum_i \psi_i(d, d) D x_i, \quad (141a)$$

$$\sum D g_{ij} dx_i \delta x_j - \sum g_{ij} \delta x_i D dx_j - \sum g_{ij} dx_i D \delta x_j = -2 \sum_i \psi_i(d, \delta) D x_i, \quad (141b)$$

$$\sum D g_{ij} \delta x_i \delta x_j - 2 \sum g_{ij} \delta x_i D \delta x_j = -2 \sum_i \psi_i(\delta, \delta) D x_i. \quad (141c)$$

It follows that $\psi_i(d, \delta)$, and in particular $\psi_i(d, d)$ and $\psi_i(\delta, \delta)$, are contragredient vectors. From (141) one computes

$$\psi_i(d, \delta) = \sum_j g_{ij} D \delta x_j + \sum_{j,k} \Gamma_{ijk} dx_j \delta x_k,$$

and from this the cogredient vector

$$p_i(d, \delta) = \sum_j g^{ij} \psi_j(d, \delta). \quad (142)$$

Setting $p(d, d) = 0$ gives, as is evident, the equations of the geodesics.

Knowledge of the cogredient vector $p(d, \delta)$ leads directly to covariant differentiation. If, for example, $h(d, \delta)$ is a form in the two rows dx and δx , then

$$\nabla h(dx, \delta x) = dh - \sum_i \frac{\partial h}{\partial dx_i} p_i(d, \delta) - \sum_i \frac{\partial h}{\partial \delta x_i} p_i(\delta, \delta) \quad (143)$$

(as written in Noether's notation) is a difference of invariants and hence itself an invariant; it is formed in such a way that second differentials cancel. Thus $\nabla h(dx, dx)$ is a form involving only first differentials: the covariant derivative of h . The same applies when h contains more rows of differentials.

The construction of the curvature form in Riemann and Lipschitz rests on the same principle. One passes from $\delta^2 f$ to the "normal form of the second variation"

$$Q = \sum D^2 g_{ij} dx_i dx_j - 2 \sum d\delta g_{ij} dx_i \delta x_j + \sum \delta^2 g_{ij} \delta x_i \delta x_j, \quad (144)$$

which, being an aggregate of invariants, is itself an invariant, and in which the third differentials have now been eliminated, generalizing the Lagrange central equation. The second differentials are eliminated in analogy with covariant differentiation by forming

$$K = Q - 2 \sum_i \{p_i(d, \delta) \psi_i(d, \delta) - p_i(\delta, d) \psi_i(d, d)\}. \quad (145)$$

This may also be written in the form

$$K = dD\psi(d, d) - 2D\psi(d, \delta)dx - 2\{p(d, \delta)\psi(d, \delta) - p(d, d)\psi(d, \delta)\}. \quad (146)$$

The law of construction of K may be phrased as follows: in $\delta^2 f$ the second differentials are eliminated by setting $p(d, d)$ and $p(d, \delta)$, equivalently the right hand sides of (141), equal to zero. This is Riemann's definition of the curvature form. It must be explicitly noted, however – and (146) makes this especially clear – that this zero substitution may be made only after the differentiation has been carried out. Lipschitz's formal construction avoids this difficulty. Correspondingly, the covariant derivative (143) may be defined by eliminating the second differentials in dh through setting the expressions $p(d, \delta)$ and the analogous ones equal to zero.

From the curvature form, by successive formation of covariant derivatives, one obtains a series of forms, Christoffel's series $[\Phi_0, \Phi_1, \dots]$. After adjoining f , their projective invariants exhaust, according to Christoffel and Ricci, the totality of the differential invariants of the quadratic form; and their equivalence with respect to linear transformations, without further integrability conditions, expresses the equivalence of two quadratic differential forms. The inner reason for this reduction theorem lies in the existence of the Riemann normal coordinates arising from the variational problem (140): they transform the geodesics through a point into straight lines and therefore transform linearly under an arbitrary transformation of the variables x . The formation of all invariants and

the equivalence question are thereby reduced to the corresponding question for the homogeneous components in the expansion of f in normal coordinates; one shows that these components are linear combinations of $f, \Phi_0, \Phi_1, \dots$. For quadratic forms this is carried out in detail by Vermeil, following a note of Noether. According to that note, the theorems and methods remain valid when an arbitrary differential expression is taken as the starting point. Thus the reach of the methods of the formal calculus of variations, as compared with those of pure elimination theory, becomes apparent.

Levi-Civita gave a geometric interpretation of setting $p(d, \delta) = 0$ as the parallel displacement of a vector. This concept, later axiomatized by Weyl, is essentially restricted to quadratic forms, but it allows a generalization of another kind. It is preserved under enlargement of the group, namely multiplication of the g_{ij} by an arbitrary function, provided that, besides the quadratic form, a linear form is also taken as given. Then $p(d, \delta)$ is uniquely determined, constructions of invariants analogous to the preceding ones can be carried out, and geodesic coordinates can be defined. The equation $p(d, d) = 0$, however, no longer arises from a variational problem. Questions of reduction theorems and equivalence have not yet been attacked in full generality in this setting; only for invariants of lowest order has Weitzenböck given a reduction theorem.

Finally, one should mention the general “invariant variational problems”, that is, those in which the simple or multiple integral J is invariant under a group in Lie’s sense. Special cases of physical interest were studied by Hamel, Herglotz, Lorentz, Fokker, Weyl, and Klein. The principled formulation by Noether shows that invariance of J under a finite group G_ρ with ρ essential parameters corresponds to ρ linearly independent divergences; invariance under an infinite group containing ρ arbitrary functions and their derivatives up to order σ corresponds to ρ identities among the Lagrange expressions and their derivatives up to order σ . In both cases the converse holds. Since the Lagrange expressions become relative invariants of the group, this gives at the same time a process for generating invariants.

K. Hentzelt, revised by E. Noether: On polynomial ideals and resultants

The following concerns the general problem of elimination theory: to determine the common zeros of all polynomials in an ideal of a polynomial ring, or equivalently the zeros of any finite number of polynomials forming a basis of the ideal. At the same time the multiplicities that occur will receive a conceptual interpretation. The coefficients of the polynomials may belong to any field; the zeros then belong to the algebraically closed field derived from it. Moreover the ideal may, without loss of generality, be assumed to be transformed, in the sense introduced below. The essential result is then the following.

To every polynomial ideal $\mathfrak{m}$ there may be attached a resultant form

$$R_{\mathfrak{m}} = R^{(1)}(x_1, \dots, x_n) \cdots R^{(i)}(x_i, \dots, x_n) \cdots R^{(n)}(x_n) \equiv 0(\mathfrak{m}), \quad (\text{A})$$

so that $R_{\mathfrak{m}}$ vanishes for all zeros of $\mathfrak{m}$ and only for them. If $\mathfrak{n}$ is a divisor of $\mathfrak{m}$ and the resultant forms $R_{\mathfrak{n}}$ and $R_{\mathfrak{m}}$ agree, then the ideals $\mathfrak{n}$ and $\mathfrak{m}$ themselves agree.

The second part of the theorem says that the resultant form gives the zeros with their characteristic multiplicity. It is analogous to the theorem that two ideals in an algebraic number field, one of which divides the other, are equal if and only if their norms agree. The inner reason for both theorems is the same.

Indeed $\mathfrak{m}$ may be considered as a module M_{n-1} of linear forms: each polynomial is regarded as a linear form in the monomials in $x_1, \dots, x_{n-1}$, and x_n is adjoined to the coefficient domain. The resultant factor $R^{(n)}$ then becomes the norm of the corresponding fundamental module G_{n-1} after M_{n-1} . This at once gives an interpretation of the multiplicity – the product of degree and exponent of a factor of $R^{(n)}$ – as the number of linearly independent residue classes. This fact was previously known only in the special case in which the resultant can be defined for indeterminate coefficients.

The derivation proceeds as follows. Sections 1 and 2 give theorems on modules of linear forms whose coefficients are either rational integers or polynomials in one indeterminate. Section 3 introduces ideals of polynomials; the isomorphism theorem of section 4 connects these ideals with the preceding module theory. Section 5 proves the second part of the main theorem. Section 7 gives the theorem about zeros, after section 6 has developed a general elimination theory. In section 6, by adjoining new indeterminates, the decomposition of the resolvent into linear forms in these indeterminates is also proved. This supplies, for the elimination method used here, an unambiguous proof of a claim made by Kronecker.

1. Modules of linear forms

In the first two sections the phrase “integral quantities” means rational integers or polynomials in one indeterminate with coefficients in an arbitrary abstract field P . By a linear form $a(\xi)$ one

understands a linear expression in finitely many indeterminates $\xi_1, \dots, \xi_k$ with integral quantities as coefficients.

A module M of linear forms is defined by the following closure conditions: together with $a(\xi)$ and $b(\xi)$ it contains $a(\xi) - b(\xi)$; together with $a(\xi)$ it contains $ca(\xi)$, where c is any integral quantity.

The rank of M is, as usual, the maximum number of linearly independent linear forms in M . The h -th determinant divisor d_h of M is the greatest common divisor of all determinants of order h formed from coefficient matrices of h forms of M . If ρ is the rank, then the non-zero determinant divisors are $d_1, \dots, d_\rho$, and the elementary divisors are

$$e_1 = d_1, \quad e_2 = d_2/d_1, \dots, \quad e_\rho = d_\rho/d_{\rho-1}.$$

The module is finite: it has a module basis consisting of exactly ρ linear forms $a_1(\xi), \dots, a_\rho(\xi)$ from which every form of M is represented linearly with integral coefficients. After an unimodular substitution $\xi = \Theta(\eta)$ the elementary divisor theorem gives the module equality

$$M = (e_1\eta_1, e_2\eta_2, \dots, e_\rho\eta_\rho). \quad (1)$$

Definition I. A module G is called a *fundamental module* if it has no proper divisor of the same rank.

Equivalently, G is a fundamental module precisely when from

$$cg(\xi) \equiv 0(G), \quad c \neq 0, \quad (2)$$

there follows $g(\xi) \equiv 0(G)$.

The relation between an arbitrary module and a fundamental module is given by the following theorem.

Theorem I. Every module M is divisible by one and only one fundamental module G of the same rank. This G is the smallest divisor of M of the same rank and, in suitable coordinates, is represented by

$$G = (\eta_1, \eta_2, \dots, \eta_\rho). \quad (3)$$

It will be called the fundamental module of M .

The assertion that G is the smallest divisor of the same rank means that G consists exactly of those linear forms $g(\xi)$ for which there is a relation

$$bg(\xi) \equiv 0(M), \quad b \neq 0. \quad (4)$$

From this description G is plainly a module. It is fundamental: if $cg(\xi) \equiv 0(G)$ with $c \neq 0$, then multiplying (4) by c gives $bcg(\xi) \equiv 0(M)$, whence again $g(\xi) \equiv 0(G)$. No second fundamental divisor of the same rank can exist, for it would have to divide G and, being fundamental, could not be a

proper divisor. The representation (3) is fundamental, since any proper divisor of it would have to contain an additional indeterminate and would therefore have higher rank.

A second concept needed later is Dedekind's quotient of two modules. If M and B are modules of linear forms, define the quotient M/B to be the set of integral quantities c satisfying

$$cB \equiv 0(M).$$

This is an ideal of integral quantities, hence a principal ideal in the present coefficient domains. Similarly, if $\mathfrak{b}$ is an ideal of integral quantities, define $M/\mathfrak{b}$ to be the set of linear forms $c(\xi)$ satisfying

$$c(\xi)\mathfrak{b} \equiv 0(M).$$

This is again a module of linear forms, and, provided $\mathfrak{b} \neq 0$, it has the same rank as M .

The quotient concept gives the reciprocal relation between a module's fundamental module and its highest elementary divisor.

Theorem II. If e_ρ is the highest elementary divisor of M and G its fundamental module, then

$$(e_\rho) = M/G, \quad G = M/(e_\rho). \quad (5)$$

Indeed, since every elementary divisor divides e_ρ , (1) and (3) give

$$e_\rho G \equiv 0(M), \quad (e_\rho)G \equiv 0(M). \quad (6)$$

Thus (e_ρ) is divisible by the quotient M/G . Conversely, from $bG \equiv 0(M)$ one obtains $b\eta_i \equiv 0(M)$ for each i , hence b is divisible by e_ρ .

For later use it is essential that the latter quotient representation of G can also be proved without recourse to the elementary divisors. This gives it greater generality.

Theorem III. If the integral quantities are polynomials in several indeterminates, then still

$$G = M/(d), \quad (7)$$

where d is any non-identically vanishing determinant of order ρ from M .

The notions of rank, fundamental module, and module quotient remain valid in this enlarged setting, although a basis representation need no longer be available. Let $a_1(\xi), \dots, a_\rho(\xi)$ be linearly independent forms of M and choose the indeterminates so that $d = |a_{ij}| \neq 0$. Then M contains special forms of the shape

$$d\xi_i + \sum_{j>\rho} b_{ij}\xi_j, \quad i = 1, \dots, \rho. \quad (8)$$

It cannot contain a non-zero form depending only on $\xi_1, \dots, \xi_\rho$, for then a determinant of order $\rho+1$ would be non-zero. Since $M/(d)$ is a divisor of the same rank as M , it is divisible by G . Conversely,

if $g \in G$, then $cg \in M$ for some $c \neq 0$; multiplying by d and using (8) shows that $dg \in M$, whence $g \in M/(d)$. This proves (7).

2. Decomposition theorems for norms and modules

We return to modules of linear forms over rational integers or over polynomials in one indeterminate over P .

Definition III. Group all linear forms congruent modulo M into residue classes. The symbol G/M denotes the system of residue classes modulo M that are represented by elements of G . The norm of G after M , written $N(G/M)$, is the determinant of the transition substitution from G to M , that is, the determinant of a substitution expressing any linearly independent module basis of M through one of the fundamental module G of M .

From (1) and (3), or from Theorem II, one obtains

$$N(G/M) = e_1 e_2 \cdots e_\rho, \quad G = M/(N(G/M)). \quad (9)$$

If the coefficients are rational integers, $N(G/M)$ is the number of residue classes of G modulo M . If the coefficients are polynomials in one indeterminate, the number of residue classes is generally infinite; the degree of $N(G/M)$ is then the number of residue classes linearly independent over P .

If B is a divisor of M of the same rank, then B contains either all representatives of a residue class or none. It is obtained by adjoining finitely many residue classes and their linear combinations to M . Consequently:

Theorem IV. If B is a divisor of M of the same rank, so that B and M have the same fundamental module G , and if

$$N(G/B) = N(G/M),$$

then $B = M$.

Theorem V. Suppose

$$N(G/M) = rs, \quad (r, s) = 1. \quad (10)$$

Then there is one and only one module R , and one and only one module S , both divisors of M of the same rank, such that

$$r = N(G/R), \quad s = N(G/S).$$

They are

$$R = M/(s), \quad S = M/(r), \quad (11)$$

and M is the least common multiple of R and S in the module sense. Moreover, if $r_i = (r, e_i)$ and $s_i = (s, e_i)$, then the elementary divisors of R and S are respectively the r_i and s_i .

The proof is the usual elementary-divisor argument. In coordinates where $M = (e_1 \eta_1, \dots, e_\rho \eta_\rho)$ and $G = (\eta_1, \dots, \eta_\rho)$, the coprimeness of r and s separates the elementary divisors into relatively prime

parts. Conversely, any modules satisfying the norm conditions have the same elementary divisors and the same fundamental module as the constructed modules; Theorem IV then forces equality.

3. Ideals of polynomials

Let $\mathfrak{m}$ be an ideal of polynomials in n indeterminates $x_1, \dots, x_n$ with coefficients in the field P . Such an ideal is, in the usual sense, a set closed under subtraction and under multiplication by arbitrary polynomials.

An ideal $\mathfrak{m}$ is called *transformed* when it remains unchanged, in the relevant sense, under a sufficiently general triangular linear transformation of the variables. By adjoining new indeterminates to the field P one may always pass to a transformed ideal. The purpose of doing so is to avoid exceptional positions in elimination: the successive resultants then have the expected degrees and the associated factors separate correctly.

The basic constructions are these. Regard the polynomials of $\mathfrak{m}$ as linear forms in the monomials in $x_1, \dots, x_i$ with coefficients in the polynomial ring $P[x_{i+1}, \dots, x_n]$. This gives a module M_i . Its fundamental module is denoted G_i . The highest elementary divisor of M_i is denoted $E^{(i+1)}$; in the elimination setting these elementary divisors are the individual resultant factors. Their product over the successive steps gives the resultant form $R_{\mathfrak{m}}$.

The highest elementary divisor is also characterized as a greatest common divisor of the maximal determinants of the coefficient matrix of the module. Thus the construction is independent of the chosen basis of the ideal. This independence is crucial: the resultant form belongs to the ideal and not to an arbitrarily selected system of generators.

For a transformed ideal, the fundamental ideals $\mathfrak{g}_i$ of the successive stages are again transformed. This is a special case of the following assertion: if $\mathfrak{m}$ is transformed, then the fundamental ideal obtained from the associated module is transformed. The proof is a direct application of the determinant criterion and of the invariance of the determinant divisors under the admissible transformations.

4. The isomorphism theorem

The relation between polynomial ideals and modules of linear forms is governed by an isomorphism theorem. Let M_i be the module obtained from $\mathfrak{m}$ by treating the polynomials as linear forms in the monomials in $x_1, \dots, x_i$, with coefficients in the remaining variables. Then the residue class system of the module agrees with the corresponding residue class system of the ideal after adjoining the remaining variables. In symbols, the construction gives an isomorphism of the form

$$G_i/M_i \simeq \mathfrak{g}_i/\mathfrak{m}$$

within the appropriate coefficient domain.

This theorem is proved by induction on i . For $i = 1$ the assertion is immediate, because one is dealing with ordinary linear forms. In the induction step one first writes the module at stage i in terms of the previous stage, then separates coefficients with respect to the new variable. The determinant divisors and the fundamental module behave functorially under this passage. Hence the quotient system of the polynomial ideal is faithfully reflected by the quotient system of the associated module.

The practical consequence is that statements about norms, elementary divisors, and residue classes of modules can be transported to polynomial ideals. Thus, when one says that a resultant factor is the norm of a fundamental module, this is simultaneously a statement about the residue classes of the corresponding polynomial ideal.

5. Norms, elementary factors, and equality of ideals

We now prove the second part of the main theorem. Let $\mathfrak{n}$ be a divisor of $\mathfrak{m}$ in the ideal sense, and suppose that the resultant forms of $\mathfrak{n}$ and $\mathfrak{m}$ agree. Passing successively to the associated modules, the equality of the resultant factors says precisely that the corresponding norms agree at each stage. By Theorem IV the modules at each stage are equal. By the isomorphism theorem the associated fundamental ideals and residue class systems are therefore equal. Running the construction back through the variables gives

$$\mathfrak{n} = \mathfrak{m}.$$

Equivalently, the resultant form does not merely locate the zero set; it records the characteristic multiplicity with which the zero set occurs. If a factor occurs with exponent a and degree d , then the product ad is the number of linearly independent residue classes represented by the corresponding component. This is the precise sense in which the resultant form supplies the multiplicities.

6. Successive elimination

The elimination theory is now developed by eliminating the variables one after another. Start from a transformed ideal $\mathfrak{a}$ and form, at each stage, the greatest common divisor of those polynomials that remain after viewing the preceding variables as coefficients. The successive greatest common divisors are denoted, in Noether's notation, by expressions such as $D^{(i)}$ and $H^{(i)}$; they are primitive in the new indeterminates and regular in the variable to be eliminated.

If a polynomial of the ideal has a zero $\alpha = (\alpha_1, \dots, \alpha_n)$, then the first elimination polynomial vanishes at the corresponding value of the eliminated variable. Conversely, because the ideal is transformed and because the determinant divisors are chosen in the general position secured by the transformation, every linear factor of the elimination polynomial extends in one and only one way to a complete zero system of the ideal after adjoining the remaining parameters.

In concrete terms, after introducing new indeterminates $u_1, \dots, u_n$ and setting

$$z = u_1x_1 + \dots + u_nx_n,$$

one obtains a resolvent polynomial whose factors have the form

$$z - (u_1\alpha_1 + \dots + u_n\alpha_n).$$

Noether's argument shows that this factorization exhausts the resolvent: no residual factor remains which is irreducible in z and the u 's. If such a factor existed, it would acquire a linear factor in an algebraic extension; the corresponding zero would have to coincide with one already obtained from the ideal, contradicting the assumption that a further factor remained.

Thus the explicit decomposition of the resolvent is

$$H(z, u) = \prod (z - (u_1\alpha_1 + \dots + u_n\alpha_n))^{e_\alpha}, \quad (36)$$

where the exponents are determined by the corresponding residue-class multiplicities. The construction differs from Kronecker's method in that it depends only on the given ideal and not on a choice of basis of that ideal.

7. Resultant form and elimination theory

To connect the resultant form with elimination theory, apply the successive elimination just described to the quotient of the ideal by the fundamental ideal of stage $n - 1$:

$$\mathfrak{a} = \mathfrak{m}/\mathfrak{g}_{n-1}.$$

First one must verify that this quotient is again transformed. Since $\mathfrak{m}$ is transformed, and since the fundamental ideals of transformed ideals are transformed, the quotient inherits the transformation property.

The determinant construction shows that the highest elementary divisor $E^{(n)}(x)$ divides the corresponding elimination polynomial $D^{(n)}(x)$. Conversely, the divisibility properties of the quotient and the regularity conditions show that $D^{(n)}(x)$ divides the resultant factor, up to the multiplicities already accounted for by the elementary divisors. The same reasoning remains valid after the auxiliary variables u are introduced, so that the factorizations of the resolvent and of the resultant agree term by term.

The conclusion is:

Theorem XIII. If the resultant form $R^{(n)}(x)$ is decomposed in an algebraic extension field of $P(u, x_{i+1}, \dots, x_n)$ into linear factors

$$x_i - \alpha_i,$$

then each α_i can be completed in one and only one way to a complete zero system

$$\alpha_1, \dots, \alpha_n$$

of $\mathfrak{m}/\mathfrak{g}_{n-1}$, and hence of $\mathfrak{m}$. The finitely many zeros of $\mathfrak{m}$ arising in this way are summarized by the explicit decomposition

$$R(z, u) = \prod (z - (u_1\alpha_1 + \dots + u_n\alpha_n))^{e_\alpha}. \quad (37)$$

This decomposition remains valid, because of the regularity of the resultant in z , after the adjoined parameters are specialized to arbitrary values in $P(u)$ or in its algebraic closure.

This also separates the zeros of $\mathfrak{m}$ according to the individual factors of the resultant form. The number of indeterminates that must be adjoined to $P(u)$ is called the dimension of the corresponding algebraic set.

Finally group, in the decomposition of $R(z, u)$, those linear factors that are conjugate over $P(u, x_{i+1}, \dots, x_n)$. One obtains a decomposition of the resultant into powers of irreducible polynomials

$$R^{(n)}(x) = S_1(x)^{\lambda_1} \dots S_r(x)^{\lambda_r}.$$

To this decomposition corresponds a representation of the relevant fundamental ideal as an intersection, or least common multiple in the ideal sense, of components. Each S_j is the norm of the corresponding component, and the exponent λ_j has the residue-class meaning already described: if γ_j is the degree of S_j , then the multiplicity $\lambda_j\gamma_j$ is the number of residue classes linearly independent over the appropriate coefficient field.

When the coefficient field P has characteristic zero, the auxiliary indeterminates u_i may be specialized to elements of P so that the required regularity of the determinants is preserved. One may even choose the specialization so that the resultant remains the greatest common divisor of finitely many maximal determinants. Then the whole elimination theory is retained after specialization; the factorization is obtained simply by substituting the chosen numerical values for the u_i in the general formula.

Algebraic and differential invariants

If one is to report on the development of algebraic and differential invariants, it is natural in both areas to restrict attention to the critical period which, according to Hilbert's remark, follows the naive and formal period. This critical period is characterized, for algebraic invariants, by Hilbert himself; for differential invariants, by Riemann. In substance, it is characterized in the first case by the arithmetic methods of algebra, which developed essentially in connection with these questions and here acquired their sharpest form; in the second, by the methods of the formal calculus of variations.

I shall therefore discuss the arithmetic methods and their significance beyond the special subject, while in the second part I shall restrict myself to the subordination of differential invariants to algebraic invariants, precisely as achieved in connection with the methods of the calculus of variations.

1. The algebraic invariant problem

The algebraic invariants are based, as is well known, on the general linear group in $x_1, \dots, x_n$:

$$x_i = \sum_j s_{ij} x'_j, \quad \text{or shortly } x = S(x'), \quad (1)$$

where the s_{ij} are indeterminates and $\Delta = |s_{ij}|$ is non-zero. If $f(a, x), g(b, x), \dots$ are forms in x of dimensions $\alpha, \beta, \dots$ with indeterminate coefficient rows $a, b, \dots$, then (1) induces transformations of $a, b, \dots$ by substituting into the forms and comparing coefficients. The transformed coefficients $a', b', \dots$ are linear homogeneous functions of $a, b, \dots$, their coefficients depending on the s_{ij} .

An invariant is a whole rational function of the indeterminates $a, b, \dots, x, y, \dots$ which, under the transformations of the variables and of the coefficient rows, reproduces itself up to a factor, namely a power of the substitution determinant:

$$J(a', b', \dots, x', y', \dots) = \Delta^\omega J(a, b, \dots, x, y, \dots). \quad (3)$$

The coefficients of J may be assumed to be rational numbers, or even integers, because every invariant with coefficients in an arbitrary number field P can be expressed linearly, with coefficients in P , through finitely many such special invariants. Likewise it is no restriction to assume J separately homogeneous in each row; any invariant is a finite sum of such components.

Conversely, the requirement (3) determines the transformations (1) and the induced transformations of the coefficients. Ostrowski and Schur have recently shown that the induced transformations can also be characterized as the totality of separate linear transformations in each row which, with certain explicit exceptions, leave a given invariant not involving the variables $x, y, \dots$ invariant.

The product of two invariants is again an invariant. A sum is an invariant if and only if the weights, the exponents ω in (3), agree. If one restricts to substitutions of determinant one, however, arbitrary

sums are again invariants, and these exhaust the invariants of the restricted group. Thus one obtains an integral domain, indeed a homogeneous integral domain: if a polynomial belongs to it, then its homogeneous components belong to it individually.

Here arises the fundamental problem at which invariant theory and arithmetic algebra as a whole developed: is this integral domain finite? That is, does it possess a finite integral basis $I_1, \dots, I_k$ such that every invariant can be represented rationally and integrally by them,

$$I = G(I_1, \dots, I_k)?$$

Hilbert's solution of the problem rests on reducing the question of an integral basis to that of an ideal basis. Gordan's proof in the binary case did not extend because of the intractable symbolic calculations, and the Mertens-Hilbert proof relied on decomposing a binary form into linear factors. I sketch Hilbert's train of thought, emphasizing the arithmetic ideas, and then mention three related questions: actual construction of the basis in finitely many steps, finiteness questions for subgroups, and finiteness questions with integrality of coefficients taken into account.

Recall the ideal definition in a finite algebraic number field. A system $\mathfrak{a}$ of numbers of the field is called an ideal if it belongs to the domain $\mathfrak{o}$ of all algebraic integers of the field; if with α and β it contains $\alpha - \beta$; and if with α it contains $A\alpha$ for every $A \in \mathfrak{o}$. Every ideal has an ideal basis,

$$\mathfrak{a} = (\alpha_1, \dots, \alpha_r),$$

meaning that every element of $\mathfrak{a}$ is of the form $A_1\alpha_1 + \dots + A_r\alpha_r$.

This definition uses only the fact that $\mathfrak{o}$ is an integral domain. It therefore carries over word for word when $\mathfrak{o}$ is replaced by any integral domain, as do the notions of ideal and ideal basis.

Hilbert's finiteness proof decomposes into three statements.

1. In the ring of all polynomials in finitely many indeterminates over a rationality domain, every ideal has a finite ideal basis.
2. A homogeneous integral domain of polynomials is finite if and only if the ideal-basis theorem holds in it.
3. For the invariant domain the ideal-basis theorem holds in the sharpened form that every ideal basis formed relative to the full polynomial ring is at the same time an ideal basis relative to the invariant domain.

In Hilbert's proof the last assertion follows from a known theorem concerning the Ω -process. E. Fischer later showed that the assertion follows already from the characteristic property of the general linear group that together with each transformation it contains the contragredient, respectively the conjugate-complex one.

2. Function fields and actual construction

The actual construction of the basis depends on a further arithmetic transformation of the finiteness concept, using function-field theory.

An integral domain $\mathcal{J}$ of polynomials is finite if and only if it contains a finite subdomain $\mathcal{H}$ such that $\mathcal{J}$ depends algebraically and integrally on $\mathcal{H}$. Any integral basis of $\mathcal{H}$ is then supplemented by a fundamental system of the algebraic integral elements to give a basis of $\mathcal{J}$.

In the case of homogeneous domains $\mathcal{J}$, such as invariant domains, for which the ideal-basis theorem holds in the sharpened form above, such a finite subdomain $\mathcal{H}$ can be obtained from finitely many polynomials $J_1, \dots, J_s$ with the property that the vanishing of these polynomials forces the vanishing of all polynomials of $\mathcal{J}$. This is again based on a general theorem of ideal theory: to every ideal of polynomials one can attach a finite set of functions that controls its zeros and hence the integral dependence of other functions over the generated subdomain.

From this upper bound K. Hentzelt obtained an upper bound for the degrees of the invariants needed for a basis. This makes the construction finite in principle, although not necessarily practical in the older symbolic sense.

3. Subgroups and integrality

The second adjoining question concerns invariants of subgroups. If one replaces the full linear group by a subgroup, then the invariant domain need not automatically satisfy Hilbert's original argument. Nevertheless, for many groups the arithmetic method persists. In particular, if the invariant domain is integral over a finite polynomial subdomain, then the same finiteness conclusion follows. This applies to finite groups and to many of the classical groups appearing in invariant theory.

The third question concerns integrality. One asks not only for rational representation of invariants by a finite basis, but for integral rational representation, with integer coefficients when the original invariant has integer coefficients. This requires replacing mere field arguments by ideal-theoretic divisibility and congruence arguments. Noether's own work on integral rational representation shows that, after suitable choice of the basic invariants and of an appropriate finite subring, integral dependence and ideal bases give the required representation. Thus the arithmetic method not only proves finiteness over fields; it also explains the integrality phenomena in invariant theory.

4. Differential invariants

I turn now to differential invariants and to the question raised at the beginning: their reduction to algebraic invariant theory.

The underlying group is, as usual, given by transformations

$$x_i = x_i(y_1, \dots, y_n). \tag{4}$$

The differentials transform by

$$dx_i = \sum_j \frac{\partial x_i}{\partial y_j} dy_j,$$

and the higher differentials transform by formulas obtained by differentiating repeatedly. If one starts with a differential expression

$$f(x, dx, d^2x, \dots, d^r x),$$

then a differential invariant is an expression that reproduces itself, possibly up to a relative factor, under the induced transformations of all the differential quantities.

The key observation is that these transformation formulas are algebraic in the derivatives of the coordinate transformation. If the derivatives of x with respect to y are treated as independent algebraic variables subject only to the non-vanishing of the Jacobian, then the problem of differential invariants becomes an algebraic invariant problem for a suitable algebraic group acting on the finite jet variables up to a prescribed order.

For each order r one obtains a finite algebraic problem. The higher differentials that occur in differentiating the invariant are eliminated by the formal variational methods described earlier. Thus the theory of differential invariants is subordinated to algebraic invariant theory: the differential invariants up to a fixed order are algebraic invariants of a finite system of forms or tensors built from the jets.

This reduction theorem shows its full force for a homogeneous differential expression $f(x, dx)$. The sequence of forms obtained from the curvature form and its covariant derivatives supplies the algebraic data whose projective invariants exhaust the differential invariants. Conversely, equivalence of the corresponding algebraic systems expresses the equivalence of the differential forms, subject to the integrability conditions already built into their construction.

Therefore the critical development of differential invariant theory parallels that of algebraic invariant theory. The old formal computations are replaced by structural, arithmetic-algebraic methods: finite generation, integral dependence, ideals, and elimination. In this way the theory of differential invariants enters the same conceptual circle as the theory of algebraic invariants.

Elimination theory and general ideal theory

Introduction

In what follows the task is to place elimination theory, in the arithmetical form that I gave to Hentzelt's exposition and that may be called an ideal theory in a polynomial domain, inside general ideal theory. At the same time the parts of the theory concerning zeros will be founded anew. The basic notions of the two theories, together with those of field theory, are collected first, so that the later argument can refer back to them.

The ordering and the new proof are grouped around four questions. First, there is an arithmetical formulation of the notion of dimension. Secondly, there is the theory of zeros. Thirdly, there is the connection between the decomposition theorems for norms and elementary divisors and the decomposition theorems of general ideal theory. Fourthly, there is the characterization of prime ideals and primary ideals by the norm and by the elementary-divisor form.

The arithmetical form of the dimension concept is given by chains of prime ideals. It is the arithmetical equivalent of the geometric fact that surfaces contain curves and curves contain points. This arithmetical formulation is shown to be identical with the parameter definition used in elimination theory. With a slight modification it can be transferred to arbitrary rings; there, together with the corresponding transfer of parts of the other questions, it gives a precise view of the structure of ideals.

For zeros Hentzelt and Noether attack the given ideal directly, as is usual, by going back to successive elimination. This does not completely avoid the character of a verification. The method followed here is the one always used for polynomials in one indeterminate. One writes the given polynomial as a product of powers of prime functions, or primary functions, and for each single prime function constructs the field of zeros as a field isomorphic to the residue-class field. The degree of the prime function gives the degree of this field; the degree of the primary function, whose zeros are the same as those of the prime function, gives the degree of the residue-class ring modulo that primary function.

Passing for each prime function to the Galois field gives the decomposition into linear factors. Thus the problem of zeros, linear factorization, and multiplicity is settled for the original polynomial. Quite analogously, for a prime ideal the system of residue classes is enlarged by forming quotients to the residue-class field, and a zero-field is constructed that is isomorphic to it. This zero-field contains a subfield generated by adjoining certain indeterminates. The degree of the norm gives the degree relative to this subfield; and by passing to the Galois field one obtains the decomposition of the elementary-divisor form, and therefore of the norm, into linear factors. For the corresponding primary ideal the zeros are the same; the decomposition is already supplied, since the norm becomes a power of the elementary-divisor form of the prime ideal. The degree of the norm gives the degree of the residue-class ring of the primary ideal relative to the subfield derived from the indeterminates.

The question for an arbitrary ideal is thereby reduced to the connection between the decomposition

theorems for norms and elementary divisors and those of general ideal theory. The fact that the zeros of an ideal are assembled from the zeros of its associated prime ideals corresponds to the decomposition of the norm into greatest primary factors. These greatest primary factors become powers of the elementary-divisor forms of the individual associated prime ideals. This gives, in general, the factorization of the norm into linear factors. The multiplicity receives its interpretation, in Hentzelt–Noether, by means of fundamental ideals and their components. The degree of a greatest primary factor of the norm is equal to the number of linearly independent residue classes, after the indicated adjunction of variables, of the fundamental ideal of level $i - 1$ modulo a component of the fundamental ideal of level i .

The fundamental ideal of level $i - 1$ will be identified with the isolated component of the decomposition that is defined by all prime ideals of dimension higher than $n - i$. A component of the fundamental ideal of level i will be identified with the isolated component defined by the prime ideal corresponding to the factor of the norm and by all prime ideals of higher dimension. The degree of the corresponding factor of the norm is then the degree of that subring of residue classes of this component which consists only of classes of the fundamental ideal of level $i - 1$.

The characterization of prime ideals and proper primary ideals is obtained as a direct consequence of the consideration of residue-class fields. If the original coefficient domain is a perfect field, then to prime ideals there correspond, as norm and elementary-divisor form, prime functions; and to proper primary ideals there correspond proper primary functions, and conversely. If the coefficient field is imperfect, one can state only necessary conditions or sufficient conditions. Examples show that the norm of a prime ideal may become properly primary, and that the elementary-divisor form of a proper primary ideal may be a prime function. More precisely, in characteristic p the norm of a prime ideal becomes the p^f -th power, $f \geq 0$, of its elementary-divisor form. For proper primary ideals whose elementary-divisor form is a prime function, the norm may become an arbitrary power, as examples again show.

Finally absolute prime ideals are considered, that is, ideals which remain prime ideals after extension of the coefficient domain to an algebraically closed field. For absolute prime ideals with coefficients in a finite algebraic number field, the characterization gives a transfer of a theorem first formulated by A. Ostrowski for absolute prime functions: the property of being a prime ideal is preserved modulo every prime ideal of the number field, with at most finitely many exceptions.

The last question is analogous to ideal theory in algebraic number fields. As the elementary-divisor form of an ideal of such a field one may take the least rational integer divisible by the ideal. For a prime ideal this is always a prime number; conversely an ideal is prime as soon as its norm is prime. The proofs run in parallel. The examples show that over an imperfect field there occur analogues of prime ideals of higher degree and of ramification ideals, whereas over a perfect field there are only prime ideals of first degree and no ramification ideals.

1. Basic notions from elimination theory, general ideal theory, and field theory

Let $\mathfrak{S}$ denote the integral domain of all polynomials in variables $y_1, \dots, y_n$ with coefficients in an abstractly given field P . The variables y are subjected to a linear transformation with indeterminate coefficients u_{rs} ,

$$y_r = \sum_s u_{rs} x_s,$$

with inverse transformation

$$x_s = \sum_r t_{sr} y_r.$$

The u_{rs} , or equivalently the t_{sr} , are adjoined to P , giving the coefficient field $P(u) = P(t)$. Let $\mathfrak{S}'$ denote the polynomial domain in $x_1, \dots, x_n$ with coefficients in this enlarged coefficient field. The domains $\mathfrak{S}$ and $\mathfrak{S}'$ are the domains underlying elimination theory. Ideals in these domains are denoted by $\mathfrak{a}, \mathfrak{b}, \dots$ and $\mathfrak{a}', \mathfrak{b}', \dots$. In any ring an ideal is defined in the usual way: together with α and β it contains $\alpha - \beta$, and together with α it contains $\lambda\alpha$ for arbitrary ring element λ . The unit ideal is denoted by $\mathfrak{o}$.

An ideal $\mathfrak{m}'$ in $\mathfrak{S}'$ is called transformed if it arises from an ideal $\mathfrak{m}$ in $\mathfrak{S}$ by the transformation and by adjoining the u 's, or equivalently the t 's. Thus $\mathfrak{m}'$ consists of the linear combinations, with coefficients in the enlarged polynomial domain, of the transformed polynomials $f(x) = f(y)$ belonging to $\mathfrak{m}$. If $f(y)$ runs through all polynomials of $\mathfrak{m}$, then $\mathfrak{m}'$ contains all corresponding $f(x)$, and because the coefficients are symmetric under the relevant changes of rows and columns, it also contains the forms obtained from them by the corresponding permutations of the variables. Unless the contrary is expressly stated, every ideal occurring below is to be assumed transformed.

If an ideal is not transformed, one may make it transformed by composing the preceding transformation with an additional linear substitution with new indeterminate coefficients. For an already transformed ideal this amounts only to replacing the old transformation coefficients by bilinear combinations of the old and new coefficients.

Throughout, expressions such as $f^{(i)}, g^{(i)}, a^{(i)}, b^{(i)}$ denote polynomials in the transformed domain that are free of $x_1, \dots, x_{i-1}$.

To each ideal $\mathfrak{m}$ there are associated n fundamental ideals

$$\mathfrak{g}_0, \mathfrak{g}_1, \dots, \mathfrak{g}_{n-1}.$$

The fundamental ideal of level $i - 1$, $\mathfrak{g}_{i-1}$, consists precisely of those polynomials $G(x)$ for which there is a non-zero polynomial $B^{(i)}$ such that

$$B^{(i)}G(x) \equiv 0 \pmod{\mathfrak{m}}.$$

The polynomial $B^{(i)}$ may depend on G , but the existence of a finite ideal basis shows that one may choose a single such $B^{(i)}$ that works for all $G(x)$ under consideration. The fundamental ideals of transformed ideals are again transformed ideals. Equivalently, $\mathfrak{g}_i$ is the ideal obtained from $\mathfrak{m}$ by

adjoining $x_{i+1}, \dots, x_n$ to the coefficient field and then restricting attention to polynomials integral in $x_1, \dots, x_i$. In particular the last fundamental ideal $\mathfrak{g}_n$ is always the unit ideal.

Every polynomial $f(x)$ can be regarded as a linear form in the monomials of $x_1, \dots, x_{i-1}$, with coefficients that are polynomials in $x_i, \dots, x_n$. In this way the ideal $\mathfrak{m}$ becomes a module M_{i-1} of linear forms over the coefficient domain of these coefficients: with two linear forms it contains their difference, and with one linear form it contains every multiple by a coefficient. The fundamental module G_{i-1} consists of the linear forms $g(\xi)$ for which some non-zero coefficient $B^{(i)}$ satisfies $B^{(i)}g(\xi) \equiv 0 \pmod{M_{i-1}}$.

The elementary divisors of these modules are understood in the usual determinantal sense. If M has rank ρ , the elementary divisors $E_1, \dots, E_\rho$ are obtained from the determinant divisors; all but finitely many of them are equal to one in the situations at hand. The highest elementary divisor $E^{(i)}$ is the greatest common divisor, in the polynomial sense, of the appropriate highest minors. It may be chosen integral and primitive in the remaining variables, and regular in the leading variable.

The product $R^{(i)}$ of the non-unit elementary divisors occurring in this module representation is called the norm of G_{i-1} modulo M_{i-1} , or the individual norm of the ideal at level i . Symbolically,

$$R^{(i)} = N(G_{i-1} \mid M_{i-1}) = N(\mathfrak{g}_{i-1} \mid \mathfrak{g}_i).$$

After adjoining the remaining variables, $R^{(i)}$ is the determinant of the transition substitution from G_{i-1} to M_{i-1} . Its degree gives the number of residue classes of $\mathfrak{g}_{i-1}$ modulo $\mathfrak{g}_i$ that are linearly independent over the relevant coefficient field.

The product of the highest elementary divisors

$$E_{\mathfrak{m}} = E^{(1)} E^{(2)} \dots E^{(n)}$$

is called the elementary-divisor form of $\mathfrak{m}$. The product

$$R_{\mathfrak{m}} = R^{(1)} R^{(2)} \dots R^{(n)}$$

is called the norm, or resultant form, of $\mathfrak{m}$. Both the elementary-divisor form and the norm are divisible by the ideal. The norm is divisible by the elementary-divisor form, and a power of the elementary-divisor form is divisible by the norm. More generally, successive products of the individual factors give congruences modulo the successive fundamental ideals, and hence finally modulo $\mathfrak{m}$.

If $\mathfrak{n}$ divides $\mathfrak{m}$ and their norms $R_{\mathfrak{n}}$ and $R_{\mathfrak{m}}$ agree, then $\mathfrak{n} = \mathfrak{m}$. Indeed, for a divisor of $\mathfrak{m}$, equality of the successive individual norms implies equality of the corresponding fundamental ideals; and since the last fundamental ideal is the unit ideal, the induction proceeds backwards. Equivalently, if $\mathfrak{n}$ is a proper divisor of $\mathfrak{m}$, then the first individual norm of $\mathfrak{n}$ that differs from the corresponding one of $\mathfrak{m}$ is a proper divisor.

A prime function means a polynomial irreducible over $P(u)$; a primary function is a power of a

prime function; a proper primary function is a power higher than the first. A decomposition of a polynomial into greatest primary factors is a decomposition in which every factor is primary and the product of any two factors is no longer primary. To the decomposition of an individual norm into greatest primary factors corresponds a representation of a fundamental ideal as a least common multiple of ideals

$$\mathfrak{g}_i = [\mathfrak{t}_1, \dots, \mathfrak{t}_s].$$

The factors of the individual norm determine the components $\mathfrak{t}_j$ of the fundamental ideals. The highest elementary divisor of such a component is the greatest common divisor of the corresponding factor and the corresponding elementary-divisor factor; it is therefore a greatest primary factor of the elementary-divisor form. These components are uniquely determined by their behavior with respect to the fundamental ideals and by the requirement that their individual norm, or their highest elementary divisor, be the prescribed greatest primary factor.

The highest dimension of an ideal is defined as follows. If $R^{(i)}$ is the first individual norm different from one, then the ideal is said to have highest dimension $n - i$. The unit ideal is assigned dimension -1 . If only one individual norm different from one occurs, the highest dimension is simply called the dimension of the ideal. If $\mathfrak{m}$ contains a non-zero polynomial free of $x_1, \dots, x_{i-1}$, then the ideal has highest dimension at most $n - i$. If $\mathfrak{m}$ has highest dimension $n - i$ and $\mathfrak{n}$ divides $\mathfrak{m}$, then $\mathfrak{n}$ has highest dimension at most $n - i$.

The zeros of an ideal are all systems of values in a suitable algebraic extension field of $P(u; x_{i+1}, \dots, x_n)$ for which every polynomial of the ideal vanishes. To each factor $R^{(i)}$ of the norm there correspond, after the indicated adjunction, as many zeros of $\mathfrak{m}$ as the degree of $R^{(i)}$ gives. Writing the factor in bilinear combinations of the transformation coefficients and further indeterminates gives an explicit decomposition, whose factors correspond to associated systems of zeros. Thus among the zeros of an ideal of highest dimension $n - i$ there are zeros depending on $n - i$ parameters, but no zeros depending on more parameters. This proves the agreement between the arithmetical dimension just defined and the usual parameter definition of dimension in elimination theory.

The ring used in general ideal theory is a commutative ring in which the finite chain condition holds: every chain of ideals

$$\mathfrak{a}_1 \supset \mathfrak{a}_2 \supset \dots$$

in which each is a proper divisor of the preceding one stops after finitely many steps. This condition is equivalent to the condition that every ideal possess a finite basis; in particular it holds in the polynomial domains just considered.

In such a ring an ideal $\mathfrak{p}$ is prime if divisibility of a product by $\mathfrak{p}$ implies divisibility of at least one factor by $\mathfrak{p}$. An ideal $\mathfrak{q}$ is primary if divisibility of a product by $\mathfrak{q}$ implies divisibility of one factor by $\mathfrak{q}$ or divisibility of a power of the other factor by $\mathfrak{q}$:

$$ab \equiv 0 \pmod{\mathfrak{p}} \Rightarrow a \equiv 0 \pmod{\mathfrak{p}} \text{ or } b \equiv 0 \pmod{\mathfrak{p}},$$

$$ab \equiv 0 \pmod{\mathfrak{q}} \Rightarrow a \equiv 0 \pmod{\mathfrak{q}} \text{ or } b^r \equiv 0 \pmod{\mathfrak{q}} \text{ for some } r.$$

To every primary ideal $\mathfrak{q}$ there is a unique associated prime ideal $\mathfrak{p}$: it divides $\mathfrak{q}$, and some power of it is divisible by $\mathfrak{q}$; moreover $\mathfrak{p}$ consists of all elements of the ring some power of which is divisible by $\mathfrak{q}$. Every prime ideal is primary; proper primary ideals are those not themselves prime.

The decomposition theorem says that every ideal admits a representation

$$\mathfrak{a} = [\mathfrak{q}_1, \dots, \mathfrak{q}_s]$$

as a least common multiple of primary ideals. It may be chosen shortest, meaning that no $\mathfrak{q}_i$ occurs in the least common multiple of the others. Then the primary components $\mathfrak{q}_i$ need not be uniquely determined, but their associated prime ideals are determined by $\mathfrak{a}$. Components belonging to isolated prime ideals, that is, to associated prime ideals which contain no other associated prime ideal, are themselves uniquely determined.

Finally, from Steinitz's field theory, recall that a field is perfect if every prime function with coefficients in it splits in a suitable extension field into distinct linear factors. Otherwise it is imperfect. Every field of characteristic zero is perfect; a field of characteristic p is perfect exactly when together with every element it contains its p -th root. Over an imperfect field a prime function can be a function of x^{p^f} and split into p^f -th powers of distinct linear factors. The integer f is the exponent. A prime function is of the first kind if the exponent is zero, i.e. if it splits into distinct linear factors in a suitable extension. An extension is of the first kind if every element is of the first kind. In a finite extension of an imperfect field there is a subfield of first kind; its degree is the reduced degree, and the exponent measures the inseparable p -power part.

2. Elementary-divisor form and norm of prime and primary ideals

We now work throughout with the polynomial domain defined above. The first lemmas state that passage to the transformed domain does not change primeness or primaryness.

Lemma I. The transform of a prime ideal is prime; the transform of a primary ideal is primary. In other words, the property of being prime or primary is preserved when the indeterminates u_{rs} are adjoined to the coefficient field.

For prime ideals the proof is based on the fact that the residue classes modulo a prime ideal form a domain without zero divisors, a property preserved by adjoining indeterminates. If $a(x)b(x)$ is divisible by the transformed ideal and if neither factor is so divisible, then, after writing the factors in terms of the inverse transformation, the highest non-vanishing coefficients with respect to a lexicographic order give a product not divisible by the original prime ideal, a contradiction.

For primary ideals the argument is parallel, but uses the Dedekind–Mertens theorem. If a is not divisible by $\mathfrak{q}$ and no power of b is divisible by $\mathfrak{q}$, one chooses transformed representatives $a(x), b(x)$ with the same property. The ideals generated by their coefficients then satisfy the Dedekind–Mertens

relation for a sufficiently high exponent; this prevents the coefficient ideal of the product from being divisible by $\mathfrak{q}$, and therefore the product itself is not divisible by the transformed ideal.

Lemma II. If

$$\mathfrak{m} = [\mathfrak{c}_1, \dots, \mathfrak{c}_s]$$

is a representation in the original polynomial domain, then the transformed ideal satisfies

$$\mathfrak{m}' = [\mathfrak{c}'_1, \dots, \mathfrak{c}'_s].$$

Consequently, in a shortest representation by greatest primary components, all components may be assumed transformed.

The proof is direct. Every transformed polynomial of $\mathfrak{m}$ is, after multiplication by a suitable denominator from the transformation coefficients, a linear combination of transformed polynomials from the original ideal. Conversely, a polynomial divisible by every transformed component is divisible by the transformed least common multiple. Applying Lemma I to a primary decomposition in the original domain gives a primary decomposition in the transformed domain.

The first characterization of prime and primary ideals by elementary-divisor form and norm is then the following.

Theorem I. The elementary-divisor form of a prime ideal is a prime function; its norm is a power of this prime function. For a primary ideal, elementary-divisor form and norm are primary functions, and indeed powers of the same prime function, namely the elementary-divisor form of the associated prime ideal. For prime and primary ideals the highest dimension agrees with the dimension in the ordinary sense, and this dimension is the same for a primary ideal and its associated prime ideal.

For the unit ideal the assertion is clear. Suppose $\mathfrak{p}$ is a prime ideal of highest dimension $n - i$. The congruences defining the fundamental ideals show that $\mathfrak{p}$ coincides with the appropriate fundamental ideal. All individual norms before level i are therefore one, and the elementary-divisor form is the single factor $E^{(i)}$. If $E^{(i)}$ split into a product of two non-units, the defining greatest-common-divisor property would imply that one factor already gives divisibility by $\mathfrak{p}$; hence $E^{(i)}$ is a prime function. The norm is a power of it because a power of the elementary-divisor form is divisible by the norm and the norm is divisible by the elementary-divisor form.

For a primary ideal $\mathfrak{q}$ the same argument gives a single elementary-divisor form at the relevant level. If $\mathfrak{p}$ is the associated prime ideal, then some power of $\mathfrak{p}$ is divisible by $\mathfrak{q}$, and $\mathfrak{q}$ divides $\mathfrak{p}$. The comparison of dimensions follows from these mutual divisibilities; the elementary-divisor form of $\mathfrak{q}$ and its norm are powers of the elementary-divisor form of $\mathfrak{p}$.

The dimension criterion for prime ideals is stated next.

Theorem II. A prime ideal has no proper divisor of the same highest dimension; conversely, every ideal with this property is prime.

The proof uses the following elementary fact.

Lemma III. If a prime ideal in a polynomial domain over a field has only finitely many residue classes linearly independent over that field, then it has no proper divisor different from the unit ideal. Equivalently, its residue-class system is already a field.

The hypothesis gives a finite number of residue classes such that every residue class is linearly dependent on them. Multiplication by a non-zero residue class is then an injective linear transformation of a finite-dimensional vector space over the ground field, hence also surjective. Thus every non-zero class has an inverse; the residue classes form a field.

To prove Theorem II, localize the situation at the stage determined by the highest dimension. A proper divisor of a prime ideal of the same highest dimension would, after adjoining the remaining parameters, give a proper divisor of a zero-dimensional prime ideal, contradicting Lemma III. Conversely, if an ideal $\mathfrak{q}$ has no proper divisor of the same highest dimension and nevertheless is not prime, choose a, b with $ab \equiv 0 \pmod{\mathfrak{q}}$ but neither factor divisible by $\mathfrak{q}$. The ideal $[\mathfrak{q}, a]$ is then a proper divisor; the dimension comparison, together with the finite-chain condition, contradicts the defining property. Hence $\mathfrak{q}$ is prime.

3. Residue-class fields and fields of zeros

Let $\mathfrak{p}$ be a prime ideal. The residue classes modulo $\mathfrak{p}$ form a domain. By forming quotients one obtains the residue-class field $(\mathfrak{R})$ of $\mathfrak{p}$. We shall construct a field of zeros isomorphic to $(\mathfrak{R})$ and use this to connect the norm with the usual splitting into linear factors.

Let $\mathfrak{p}$ be of dimension $n - i$. The residue classes of the last $n - i$ variables may be taken as parameters. After adjoining them, the remaining residue classes are algebraic over the field $P(u, x_{i+1}, \dots, x_n)$. The elementary-divisor form $P^{(i)}$ is the prime function of a suitable class, say the class of a fundamental linear form. The residue class of this form is algebraic over the parameter field and generates, together with the parameters, a field isomorphic to the quotient field of the residue-class domain.

Theorem III. To the residue-class field of a prime ideal $\mathfrak{p}$ there corresponds an isomorphic field of zeros. If $\mathfrak{p}$ has dimension $n - i$, this zero-field is a finite algebraic extension of the parameter field $P(u, x_{i+1}, \dots, x_n)$. Conversely every zero-field of $\mathfrak{p}$ of this transcendence degree is isomorphic to the residue-class field.

A zero $\alpha = (\alpha_1, \dots, \alpha_n)$ is obtained by sending each residue class (x_j) to the corresponding element α_j in an extension field. Since exactly the congruences of $\mathfrak{p}$ become zero, this gives an isomorphism of the residue-class field with the field $P(\alpha_1, \dots, \alpha_n)$. Conversely any zero-field gives such a residue-class homomorphism and hence the same abstract field.

Passing from the zero-field to the corresponding Galois field over the parameter field gives the factorization of the elementary-divisor form. If α_1 is a chosen primitive element, then the associated prime function in α_1 splits in the Galois field into linear factors. Returning from the transformed

variables x to the original variables y , the element α_1 becomes the fundamental form

$$u_1\beta_1 + \cdots + u_n\beta_n,$$

where $\beta = (\beta_1, \dots, \beta_n)$ runs through systems of zeros in the original variables depending algebraically on the parameters. Thus the linear factors of the norm are precisely the factors corresponding to the conjugate systems of zeros.

In modern terms, the residue-class field is the function field of the irreducible component determined by $\mathfrak{p}$, and the zero-field is a concrete algebraic realization of that function field in which the coordinates of the generic point are displayed. Noether's proof keeps the older elimination terminology, but the construction is this one.

Lemma IV. The elementary-divisor form of a prime ideal is the prime function attached to a primitive element of its residue-class field; its norm is the product of the elementary divisors obtained in the module representation, and hence a power of that prime function. If the coefficient field is perfect, the power is the first power.

In an imperfect field the prime function may be inseparable. Then the norm records the full degree, while the elementary-divisor form records the reduced degree. This is the source of the possible power in Theorem I.

Theorem IV. In the Galois field determined by a prime ideal $\mathfrak{p}$, the norm $N(\mathfrak{p})$ decomposes into linear factors corresponding to the conjugate zeros of $\mathfrak{p}$. The degree of the norm equals the number of residue classes linearly independent over the parameter field, counted with the inseparable exponent when the coefficient field is imperfect.

For a primary ideal $\mathfrak{q}$ associated with $\mathfrak{p}$ the zero-field is the same as that of $\mathfrak{p}$, because $\mathfrak{p}$ and $\mathfrak{q}$ have the same zeros. The norm of $\mathfrak{q}$ is a power of the elementary-divisor form of $\mathfrak{p}$; the exponent measures the length of the residue-class ring of $\mathfrak{q}$ over the residue-class field of $\mathfrak{p}$ at the corresponding level.

4. Dimension

We now compare the arithmetical dimension defined by the first non-unit norm with the usual parameter dimension.

Theorem V. The dimension of a prime ideal is given by the transcendence degree of its residue-class field over $P(u)$. If $\mathfrak{p}$ is of dimension $n - i$, then its residue-class field contains $n - i$ algebraically independent parameters and is algebraic over the field generated by these parameters.

The proof follows from the construction of the zero-field. The last $n - i$ variables can be chosen so that their residue classes are algebraically independent, while every further residue class is algebraic over them. No set of $n - i + 1$ residue classes can be algebraically independent, because the relevant elementary-divisor form gives an algebraic equation. Conversely the selected parameter classes are independent because otherwise a non-zero polynomial in them would belong to $\mathfrak{p}$, forcing a non-unit individual norm at an earlier level.

This proves the equivalence of the dimension in elimination theory with the arithmetical chain definition. Noether's formulation is deliberately independent of geometric language: an ideal has dimension $n - i$ when the first non-trivial elimination norm occurs at level i ; it has the same dimension when its residue-class field has transcendence degree $n - i$; and the same number is characterized by the maximum length of chains of prime ideals lying above it.

Theorem VI. Let

$$\mathfrak{p}_0 \supset \mathfrak{p}_1 \supset \cdots \supset \mathfrak{p}_r$$

be a chain of prime ideals. Then the dimensions strictly decrease along a proper chain. Conversely, a prime ideal of dimension $n - i$ can be placed in a chain of prime ideals of length $n - i$ ending at a maximal ideal. Thus the dimension equals the maximum length of a chain from the given prime ideal down to a maximal ideal, counted in the appropriate direction.

The geometric translation is that passing from a variety to a proper subvariety lowers the number of independent parameters. In the polynomial domain, the arithmetical proof uses divisibility of norms: a proper divisor of a prime ideal cannot have the same highest dimension by Theorem II; hence the dimension must decrease. Conversely, the existence of parameters and of algebraic equations in the remaining variables allows one successively to specialize a parameter and obtain a proper prime divisor at each step until a zero-dimensional prime ideal is reached.

Theorem VII. A prime ideal $\mathfrak{p}$ is of dimension $n - i$ if and only if it contains a prime chain of exactly i algebraic elimination steps, or equivalently if the first non-unit individual norm attached to $\mathfrak{p}$ occurs at level i .

This theorem supplies the promised arithmetical formulation of the dimension concept. It is invariant under the transformations used in elimination theory and does not depend on a distinguished basis of the ideal.

5. Connection with the decomposition theorems of general ideal theory

The components introduced in Hentzelt–Noether through fundamental ideals are now identified with the isolated components of the primary decomposition of general ideal theory. This is the main comparison theorem.

Let the associated prime ideals of $\mathfrak{m}$ be arranged by dimension:

$$\mathfrak{p}_{0,1}, \dots; \quad \mathfrak{p}_{1,1}, \dots; \quad \mathfrak{p}_{2,1}, \dots; \quad \dots,$$

where $\mathfrak{p}_{\lambda,\mu}$ denotes a prime ideal of dimension $n - \lambda$. Let $\mathfrak{q}_{\lambda,\mu}$ be the corresponding primary ideals in a fixed primary decomposition of $\mathfrak{m}$.

For a given level i , the fundamental ideal of level i is, by definition, the unit ideal for those primary components whose associated prime ideals have dimension less than the corresponding threshold; for components of higher dimension it remains the given primary component. Thus the fundamental

ideals are least common multiples of precisely those primary components whose associated primes are sufficiently large in dimension.

Lemma V. The fundamental ideal of level i is the isolated component of the primary decomposition defined by all associated prime ideals of dimension greater than $n - i$. More precisely, it is the least common multiple of those primary components whose associated prime ideals have dimension greater than $n - i$.

This follows from the defining congruence for the fundamental ideals. Multiplication by a suitable polynomial in the eliminated variables removes the components in smaller dimensions, while components of higher dimension remain. Since isolated components are uniquely determined in general ideal theory, the fundamental ideals are independent of the chosen primary decomposition.

Theorem VIII. If all associated prime ideals of $\mathfrak{m}$ have the same dimension $n - i$, then the norm of $\mathfrak{m}$ decomposes into greatest primary factors corresponding one-to-one to the associated prime ideals. Each factor is a power of the elementary-divisor form of the corresponding prime ideal.

In this unmixed case the relevant fundamental ideals reduce directly to the primary components. The individual norm at level i is the only non-unit norm. Its factorization into greatest primary factors therefore mirrors the primary decomposition of $\mathfrak{m}$.

Theorem IX. In the general case, where associated prime ideals of different dimensions occur, the decomposition of the norm of $\mathfrak{m}$ is obtained by grouping the contributions according to dimension. The greatest primary factors of the norm at the first non-trivial level correspond to the isolated components of greatest dimension; after their removal the next individual norm gives the components of the next dimension, and so on.

This is precisely the arithmetical form of the statement that the zeros of an ideal are the union of the zeros of its associated prime ideals. The norm does not merely vanish at those zeros; its factorization displays the associated components in the order of their dimension.

Theorem X. The degree of a greatest primary factor of the norm corresponding to an associated prime ideal $\mathfrak{p}$ is equal to the number of residue classes, linearly independent over the parameter field, of the component of the fundamental ideal at level $i - 1$ modulo the component of the fundamental ideal at level i determined by $\mathfrak{p}$.

This theorem gives the multiplicity. Instead of assigning multiplicity by a geometric convention at the zeros, Noether assigns it by a length, or number of independent residue classes, in a quotient of isolated components. For polynomials in one indeterminate this is exactly the ordinary multiplicity of a root: the primary component is the power of the corresponding prime ideal, and the number of residue classes is the exponent. In higher-dimensional elimination theory, the same algebraic length replaces the root multiplicity.

The theorem also explains the role of the “basic ideals” in Hentzelt–Noether. They are not additional arbitrary constructions; they are the isolated components selected by dimension in the primary decomposition. Their quotient lengths are the multiplicities recorded by the norm.

Theorem XI. The decomposition of an ideal into primary components and the decomposition of its norm into greatest primary factors determine each other at the level of associated prime ideals and multiplicities. In particular, two ideals one of which divides the other and which have the same norm must be equal.

This recovers, in the polynomial domain, the arithmetical theorem analogous to the theorem on norms of ideals in algebraic number fields.

6. Characterization of prime ideals and proper primary ideals by elementary-divisor form and norm

The preceding theorems characterized primary ideals in a way that did not yet separate prime ideals from proper primary ideals. We now obtain the separation.

Theorem XII. A necessary condition for an ideal to be prime is that its elementary-divisor form be a prime function; a sufficient condition is that its norm be a prime function. Equivalently, the elementary-divisor form of a prime ideal is always a prime function, and the norm of a proper primary ideal is always a proper primary function.

The first assertion has already been proved in Theorem I. For the sufficient condition, suppose the norm of an ideal is a prime function. If the ideal had a proper divisor of the same highest dimension, then the first individual norm in which the two ideals differ would be a proper divisor of the given prime function, impossible unless it is a unit. Thus every proper divisor has lower highest dimension, and Theorem II shows that the ideal is prime.

A second proof uses the following lemma.

Lemma VII. The residue-class system represented by polynomials modulo the elementary-divisor form $E^{(i)}$ of an ideal whose associated prime ideals all have dimension $n - i$ is isomorphic to a subsystem of the residue-class system modulo the ideal. The degree of the elementary-divisor form is the degree of this subsystem over the quotient field $P(u, x_{i+1}, \dots, x_n)$.

Indeed, classes represented by the same polynomial correspond to each other. The correspondence is injective because exactly those polynomials divisible by the elementary-divisor form represent the zero class in the subsystem. If the norm is a prime function and hence agrees in degree with the elementary-divisor form, this subsystem exhausts all residue classes. Since a residue-class system modulo a prime function has no zero divisors, the ideal is prime.

In general Theorem XII cannot be sharpened to necessary and sufficient conditions. Over imperfect fields examples show two failures: the norm of a prime ideal can be a proper power, and the elementary-divisor form of a proper primary ideal can be a prime function. However the perfect-field case is clean.

Theorem XIII. If the coefficient field P is perfect, then the norm and the elementary-divisor form of a prime ideal are prime functions and are identical. The norm and the elementary-divisor form of

a proper primary ideal are proper primary functions. Thus, over a perfect field, the condition that the elementary-divisor form be a prime function is necessary and sufficient for primeness; the same condition may be stated using the norm.

It remains only to show that over a perfect field the ideal is prime whenever the elementary-divisor form is a prime function, and that the norm then equals this prime function. Since P is perfect, a prime elementary-divisor form is of first kind. The residue-class field obtained by adjoining a root is therefore separable; by the usual Lagrange formula all the necessary residue classes lie in the subsystem described in Lemma VII. That subsystem exhausts the residue classes of the ideal. It has no zero divisors, so the ideal is prime. The equality of degrees gives equality of norm and elementary-divisor form.

For imperfect coefficient fields the following sharper statements hold.

Theorem XIV. For a prime ideal over a field of characteristic p , the norm is a p^f -th power of its elementary-divisor form, where f is the exponent of the finite extension of residue-class fields. If the elementary-divisor form of a proper primary ideal is a prime function, then its norm is a power of that prime function; the exponent need not be a power of p in general.

The proof follows the field-theoretic decomposition into reduced degree and inseparable exponent. The elementary-divisor form measures the reduced degree; the norm measures the full degree. Their quotient is therefore the inseparable p^f part for prime ideals. For proper primary ideals the nilpotent, or primary, multiplicity contributes an additional exponent.

Noether gives explicit examples in characteristic 2 and 3. If P is obtained by adjoining independent indeterminates to a finite field, then an ideal such as

$$(y_1^2 + \lambda, y_2^2 + \mu)$$

can be prime, because its residue-class field is purely inseparable of degree four; its elementary-divisor form has reduced degree while its norm is the square or fourth power accordingly. Specializing parameters can produce a proper primary ideal whose elementary-divisor form is still prime. In characteristic 3, a similar construction shows that the norm exponent for a proper primary ideal need not equal the characteristic exponent predicted for prime ideals.

These examples demonstrate exactly where imperfectness enters. In a perfect coefficient field, no inseparable power is hidden in the field extension, and the elementary-divisor form and norm detect primeness without ambiguity.

7. Absolute prime ideals

A prime ideal with coefficients in a field P is called an absolute prime ideal if it remains prime after extending the coefficient field to the algebraically closed field A belonging to P . Correspondingly, a prime function is absolute if it remains irreducible, or prime, over A . The characterization is the following.

Theorem XV. A necessary and sufficient condition for a prime ideal to be absolute is that its elementary-divisor form, chosen integral and primitive in the transformation coefficients u , be an absolute prime function as a polynomial in the x and u . The elementary-divisor form then coincides with the norm, so the condition may also be stated for the norm.

The proof starts from the observation that norm and elementary-divisor form are preserved under algebraic extension of the coefficient field. Each individual factor is defined as a greatest common divisor in a polynomial ring, and this property is unchanged by algebraic extension. Passing from P to the algebraically closed field A passes in particular to a perfect field. By Theorem XIII, over A primeness is equivalent to the elementary-divisor form being a prime function. Hence the original prime ideal is absolute exactly when that form is absolutely prime. Since over A the norm equals the elementary-divisor form, and equality descends back to P by the preceding invariance, the assertion follows.

For absolute prime functions with coefficients in a finite algebraic number field, Ostrowski proved that the function remains absolute prime modulo every prime ideal of the number field with only finitely many exceptions. The transfer to ideals uses the following finite-denominator statement.

Theorem XVI. Let the coefficient domain be the ring $\mathfrak{o}^*$ of algebraic integers in a finite algebraic number field, rather than a field. Then the norm and elementary-divisor form of a polynomial ideal remain the norm and elementary-divisor form after reduction modulo every prime ideal of $\mathfrak{o}^*$, with at most finitely many exceptions.

The proof modifies the polynomial domain. Its coefficients now lie in $\mathfrak{o}^*$, while the transformation coefficients u are adjoined polynomially. In forming the module representation for the individual norms, one reduces powers of the eliminated variable modulo a polynomial regular in that variable. The possible denominators that occur are products of finitely many polynomials in the u with coefficients in $\mathfrak{o}^*$. Their coefficients generate an ideal of $\mathfrak{o}^*$ divisible by only finitely many prime ideals. Avoiding these prime ideals preserves the module representation after reduction.

The same is done for the greatest-common-divisor property defining the norm and elementary-divisor form. One chooses the norm and elementary-divisor form so that they are divisible by the ideal over $\mathfrak{o}^*[u]$. For all but finitely many prime ideals of $\mathfrak{o}^*$ the relevant ranks of the modules, the relevant determinants, and the greatest-common-divisor relations survive reduction. Thus, modulo such a prime ideal, the reduced forms are still exactly the norm and elementary-divisor form of the reduced ideal.

Combining Theorems XV and XVI with Ostrowski's theorem gives the final result.

Theorem XVII. If $\mathfrak{p}$ is an absolute prime ideal with coefficients that are algebraic integers in a finite algebraic number field, then $\mathfrak{p}$ remains a prime ideal, more precisely an absolute prime ideal, modulo every prime ideal of the number field, with at most finitely many exceptions.

Choose the norm $R_{\mathfrak{p}}$ as in Theorem XVI so that it is divisible by $\mathfrak{p}$ over $\mathfrak{o}^*[u]$. Then

$$R_{\mathfrak{p}} = T(u)P(x),$$

where $P(x)$ may be chosen integral and primitive in the u . By Theorem XV, regarded as a polynomial in u and x with coefficients in the number field, P is an absolute prime function. Ostrowski's theorem says that P remains absolute prime modulo all but finitely many prime ideals. Avoiding also the finite exceptional set in Theorem XVI, this reduced P is the norm of the reduced ideal. Theorem XV then implies that the reduced ideal is again an absolute prime ideal.

This completes the transfer of the arithmetic theorem on absolute prime functions to absolute prime ideals.

Elimination theory and ideal theory

I want to show how elimination theory can be built up in complete parallel with the theory of zeros for polynomials in one indeterminate. This construction rests on a connection between the elimination theory given by Hentzelt and the methods of field theory as developed by Steinitz, and of ideal theory as I have developed it.

First consider the case of polynomials in one indeterminate. The coefficient domain is a fixed field P , and notions such as prime function are to be understood relative to P . The field may be abstractly defined in Steinitz's sense; in special cases it may be the field of rational numbers or the field of all complex numbers, as was formerly usual in elimination theory. For polynomials in one indeterminate the theory of zeros falls into four steps.

First comes the decomposition theorem, which reduces the question to prime functions, that is, to polynomials irreducible over P . From a factorization

$$a(x) = q_1(x)q_2(x) \cdots q_s(x),$$

where the underlying $p_i(x)$ are distinct prime functions and the $q_i(x)$ are their powers, hence greatest primary functions, it follows that $a(x)$ vanishes exactly at the zeros of the prime functions $p_i(x)$.

Second, one constructs the zero-field for a prime function $p(x)$. Such a field

$$K = P(\alpha), \quad p(\alpha) = 0,$$

can always be constructed, by Steinitz, as an extension field of P isomorphic to the residue-class field of $p(x)$. Conversely, every zero-field lying in an extension field of P is isomorphic to that residue-class field. Thus zero-field and residue-class field are abstractly identical.

Third, one decomposes $p(x)$ into linear factors in the Galois field determined by $p(x)$. Repetition of the construction just described leads to an essentially uniquely determined Galois field $P(\alpha_1, \dots, \alpha_r)$ large enough for $p(x)$ to split into linear factors.

Fourth, one interprets multiplicity for the original polynomial $a(x)$. By the preceding steps the zeros and the corresponding linear factorization are known for an arbitrary polynomial. The multiplicity may be taken to be the degree of the corresponding primary function $q(x)$, equivalently the number of residue classes modulo $q(x)$ that are linearly independent over P . If P is algebraically closed and the prime functions are linear, this number is exactly the usual multiplicity of the zero.

Now pass to general elimination theory. Let $\mathfrak{S}$ be the domain of all polynomials in $x_1, \dots, x_n$ with coefficients in P . Elimination theory may then be defined as the theory of the zeros of an ideal in this polynomial domain. An ideal $\mathfrak{a}$ is a system of polynomials such that with f and g it contains $f - g$, and with f it contains Af for every polynomial A in $\mathfrak{S}$. A zero of $\mathfrak{a}$ is a system of values $\alpha_1, \dots, \alpha_n$ in an extension field of P for which every polynomial f in $\mathfrak{a}$ vanishes.

If $f_1, \dots, f_s$ is an ideal basis, so that every $f \in \mathfrak{a}$ has the form

$$f = A_1 f_1 + \dots + A_s f_s,$$

then the zeros of $\mathfrak{a}$ are exactly the common zeros of $f_1, \dots, f_s$. Conversely, any finite system of polynomials may be regarded as a basis of the ideal it generates. Thus the ideal-theoretic definition of zeros is the usual one, but avoids the arbitrariness of singling out a particular basis. In one variable this difficulty did not appear, because only principal ideals occur and the basis polynomial is determined up to a unit. Even there, however, the ideal viewpoint gives a clearer interpretation.

The theory again falls into four steps.

I. The decomposition theorem represents an ideal as a least common multiple of greatest primary components and thereby reduces the question to prime ideals. An ideal $\mathfrak{p}$ is prime if divisibility of a product by $\mathfrak{p}$ implies divisibility of one factor; an ideal $\mathfrak{q}$ is primary if divisibility of a product by $\mathfrak{q}$ implies divisibility of one factor or of some power of the other factor. To every primary ideal $\mathfrak{q}$ there belongs a unique associated prime ideal $\mathfrak{p}$ which divides $\mathfrak{q}$ and some power of which is divisible by $\mathfrak{q}$. One has a representation

$$\mathfrak{a} = [\mathfrak{q}_1, \dots, \mathfrak{q}_s],$$

where the $\mathfrak{q}_i$ are greatest primary components. Although the primary components themselves need not be uniquely determined, their associated prime ideals are uniquely determined by $\mathfrak{a}$. Since every associated prime ideal divides $\mathfrak{a}$, and conversely a product of powers of the associated primes is divisible by $\mathfrak{a}$, the zeros of the ideal agree with the zeros of its associated prime ideals.

II. The residue classes modulo a prime ideal form a ring without zero divisors. If the prime ideal is not the unit ideal, this ring contains a non-zero element, and by adjoining quotients it becomes the residue-class field of the prime ideal. There is always an extension field of P isomorphic to this residue-class field; it becomes the zero-field of the prime ideal. Thus

$$K = P(\alpha_1, \dots, \alpha_n),$$

where $(\alpha_1, \dots, \alpha_n)$ is a zero of $\mathfrak{p}$. The field has transcendence degree k , $0 \leq k < n$, over P : it contains k algebraically independent elements over P , while every $k + 1$ elements are algebraically dependent. Conversely, every zero-field of transcendence degree k is isomorphic to the residue-class field. Therefore zero-field and residue-class field are abstractly the same object.

III. The norm of $\mathfrak{p}$ is decomposed into linear factors in the Galois field determined by $\mathfrak{p}$. If the variables $x_1, \dots, x_n$ are obtained from original variables $y_1, \dots, y_n$ by a linear transformation with indeterminate coefficients, then the zero-field of transcendence degree k can be viewed as a finite algebraic extension of $P(x_{k+1}, \dots, x_n)$. Passing to its Galois field, the prime function for a primitive element splits into linear factors. This prime function plays, after returning to the y variables, the role of the fundamental equation: its root is the fundamental linear form in a zero system β . The factorization therefore gives the product over all zeros. The prime function, or a power of it,

is also the norm $N(\mathfrak{p})$ of the prime ideal, obtained by regarding $\mathfrak{p}$ as a module of linear forms in the monomials in the eliminated variables and defining the norm as the product of the elementary divisors. Over a perfect coefficient field this norm is exactly the prime function; over an imperfect field it may be a power.

IV. For the original ideal $\mathfrak{a}$, one regards it successively as a module of linear forms in the monomials of $x_1, \dots, x_{i-1}$ for $i = 1, \dots, n$. At each step, with coefficient field $P(x_i, \dots, x_n)$, only finitely many elementary divisors differ from one. Their product gives the individual norm, and the norm $N(\mathfrak{a})$ is the product of all individual norms. The greatest primary factors of this norm correspond one-to-one to the associated prime ideals $\mathfrak{p}$ of $\mathfrak{a}$. Each such factor is a power of the norm of $\mathfrak{p}$ or, in the inseparable case, of its highest elementary divisor. Thus the norm of $\mathfrak{a}$ splits according to the associated primes, and the preceding steps give the linear factorization for all zeros of $\mathfrak{a}$.

The multiplicity attached to the zeros corresponding to an associated prime ideal $\mathfrak{p}$ is the degree of the corresponding primary factor of the norm. This degree is the number of linearly independent residue classes, over the appropriate parameter field, of the isolated component determined by $\mathfrak{p}$ and by the prime ideals of higher dimension. In the one-variable case this is exactly the degree of the primary factor and therefore the usual multiplicity.

The case of one indeterminate is thus fully included by passing from a polynomial to its principal ideal. Its factorization may be read either as a least common multiple of primary components or as a decomposition of the norm into norms of prime ideals. The zero-field belongs to the prime ideal defined by $p(x)$, while the norm gives the linear factorization. The multiplicity is the number of residue classes in the relevant quotient, and the lowest fundamental ideal is the unit ideal whereas the first fundamental ideal is already the ideal itself.

In this form the usual elimination theory is incorporated in ideal theory: the norm of the ideal has the meaning of a resultant. The old elimination problem becomes the part of ideal theory in the polynomial domain that treats, alongside ideal decompositions, the decomposition of norms and the relation between the two decompositions.

Abstract construction of ideal theory in algebraic number fields

The following abstract properties were indicated as completely equivalent to the ideal theory in an algebraic number field, and hence as characterizing all rings that possess such an ideal theory. They are domains with identity element and without zero divisors in which the double chain condition holds: every divisor chain of ideals stops after finitely many steps, and so does every multiple chain modulo every ideal different from the zero ideal. In addition the ring is integrally closed in its quotient field. If the last assumption is omitted, one obtains theorems on ideal theory in a finite order, some of which are already found without proof in Dedekind's third edition. A detailed exposition will appear in the *Mathematische Annalen*.

Hilbert numbers in ideal theory

By Hilbert numbers one generally understands numbers that concern the enumeration of residue classes of ideals. Hilbert himself gave, in his "characteristic function", a function which for ideals in a polynomial domain gives, from a certain bound on, the exact number of linearly independent residue classes. For lower degrees Macaulay supplemented this function by a generating function, which Ostrowski could evaluate explicitly.

Macaulay also introduced numbers of another kind. These have proved most essential for general ideal theory because they admit an abstract formulation. In accordance with the general decomposition theorems it suffices to define them for primary ideals $\mathfrak{q}$. They count the number of linearly independent residue classes with respect to the residue-class field of the associated prime ideal $\mathfrak{p}$.

These numbers decompose into subsystems given by the numbers of irreducible components of

$$\mathfrak{q}, \quad \mathfrak{q}/\mathfrak{p}, \quad \mathfrak{q}/\mathfrak{p}^2, \quad \dots$$

The number of the i -th subsystem is also characterized as the number of terms in a composition series running from $\mathfrak{q}/\mathfrak{p}^i$ to $\mathfrak{q}/\mathfrak{p}^{i-1}$. The total composition series and its Hilbert numbers form the substitute for the representation of primary ideals as powers of prime ideals, a representation which is no longer valid in general, for example in finite orders of an algebraic number field.

Group characters and ideal theory

Let G be a finite group and let K be a splitting field for G . The group algebra $K[G]$ decomposes into simple components corresponding to the irreducible characters. The arithmetical questions concern what happens when one replaces K by a subring, or by the ring of algebraic integers in a number field, and considers ideals or modules stable under the group operation.

The character equations give idempotent decompositions after extension to the splitting field. Ideal theory supplies the descent from these decompositions to integral modules. In this way the

elementary divisors attached to a group representation can be read in parallel with the elementary divisors of ideals. The irreducible characters describe the simple constituents over the field; the corresponding integral ideal theory describes how these constituents combine when denominators and divisibility are taken into account.

This viewpoint places the arithmetic of group characters beside the arithmetic of polynomial ideals. In both cases a decomposition over a field gives a formal splitting, while ideal theory records the integral and modular data lost in the field decomposition alone. The connection is especially useful when studying representations modulo prime ideals of the coefficient ring: the character-theoretic decomposition may remain valid, refine, or coalesce according to the behavior of the associated ideals after reduction.

Abstract construction of ideal theory in algebraic number and function fields: first installment

In the following an abstract characterization is given of all rings whose ideal theory agrees with the ideal theory of all algebraic integers in an algebraic number field, that is, rings in which every ideal is uniquely expressible as a product of powers of prime ideals. Among the best known examples are also the ring of all integral elements of an algebraic function field in one indeterminate, and more generally, for several indeterminates, the rings obtained by restricting oneself, in Kronecker's sense, to ideals of highest dimension and passing to the corresponding quotient ring, the functional domain.

The axioms completely equivalent to this ideal theory, for the commutative ring under consideration, are the following.

I. Divisor chain condition. Every chain of ideals in which each ideal is a proper divisor of the preceding one terminates after finitely many steps; equivalently, in every divisor chain of ideals there is an index from which on all ideals are equal.

II. Multiple chain condition modulo each non-zero ideal. Every chain of ideals, all of which divide a fixed ideal different from the zero ideal, and in which each ideal is a proper multiple of the preceding one, terminates after finitely many steps.

III. There is an identity element for multiplication.

IV. The ring has no zero divisors.

V. Integral closedness in the quotient field. Every element of the quotient field integral over the ring belongs to the ring.

From these axioms Noether develops step by step an increasingly restricted ideal theory, ending with the desired theory, and conversely shows that all five axioms are actually satisfied in the classical examples.

From the divisor chain condition I follows the representation of an ideal as a least common multiple of finitely many primary ideals belonging to distinct prime ideals. Noether repeats the proof because it is simplified by using the ideal quotient and because a gap in the earlier proof is to be corrected: the old proof used, at one point, the existence of a multiplicative identity, which had not been assumed.

The multiple chain condition II implies that in the residue-class ring modulo every ideal different from the zero ideal, a prime ideal can have no proper divisor other than the unit ideal containing all elements. Hence the primary components of such ideals are uniquely determined except for components belonging to the unit ideal. In a somewhat less sharp form this result already occurs in work of Masazo Sono. Sono's hypothesis of the existence of a composition series in the residue-class ring is equivalent to the double chain condition in that ring. By the double chain condition Noether

means the simultaneous validity of the divisor chain condition and the multiple chain condition, without restricting to non-zero ideals.

If axiom III, the existence of the unit element, is added, then the unit ideal does not occur as an associated prime ideal. The primary components are therefore uniquely determined; they are pairwise relatively prime, and their least common multiple is their product. Axioms I, II, and III hold for all finite orders of an algebraic number field. Dedekind had already stated, without a complete proof, the unique representation of an ideal as a product of pairwise relatively prime one-type ideals, i.e. primary components.

Axiom IV, the absence of zero divisors, implies that the different powers of an ideal different from both the zero ideal and the unit ideal are all distinct, and that the quotient field exists. If finally axiom V is added, integral closedness in the quotient field, then the primary ideals, which by IV are uniquely determined, become powers of prime ideals. This gives the usual decomposition theorem. The proof uses a consequence of integral closedness already drawn by Dedekind in the number-field case: a genuinely fractional element can be written as a quotient of integral elements in such a way that even the square of the numerator is not divisible by the denominator.

In later treatments this consequence is often replaced by the less transparent generalized Gauss theorem, or by corresponding module theorems that give somewhat more. Those results require basis elements in their application; the proof here works with ideals themselves.

In the converse direction the axioms other than V follow almost directly from unique factorization of ideals. Axiom V follows from the proof that every genuinely fractional element can be represented so that no power of the numerator is divisible by the denominator. That is even stronger than Dedekind's original consequence of integral closedness. The proof is obtained only after drawing the customary conclusions from the ideal factorization: principal-ideal representation modulo a fixed ideal, and the theory of fractional ideals, i.e. ideal quotients in the field. Dedekind had already known that integral closedness follows from the representation by prime-ideal powers.

The axioms I–V imply that four decompositions which are generally distinct coincide: decomposition into least pairwise relatively prime ideals, into mutually prime ideals, into greatest primary components, and into irreducible ideals. Conversely, this coincidence together with axioms I–IV implies integral closedness. If zero divisors are allowed in the ring, then the remaining axioms, with V interpreted as integral closedness in the total quotient ring, do not necessarily imply this coincidence; the residue-class ring modulo certain ideals of a finite order gives examples.

Section 1 sketches the theory of elements integral over a ring, through Dedekind's consequences of integral closedness. Section 2 proves the transfer of chain conditions from a ring to modules of linear forms over it. Section 3 applies this to passage to finite extension fields of first kind: axioms I–IV are preserved for every order, and axiom V holds for the order consisting of all integral elements. This places both number fields and function fields under the same scheme. In particular, once it is shown directly that the functional domain derived from integral polynomials in several indeterminates satisfies the axioms, Kronecker's ideal theory is fully incorporated. The later ideal theory does not

depend on these classificatory sections; sections 4 and 5 give the foundations independent of axioms I–V and show more clearly which parts of the theory do not require finiteness hypotheses.

1. Theory of integral elements

Let T be a commutative ring without zero divisors and with multiplicative identity e , and let R be a fixed subring of T containing e . The words module, order, and integral are throughout to be understood relative to this fixed ring R .

A system M of elements of T is called an R -module, or simply a module, if with two elements α and β it contains $\alpha - \beta$, and with α it contains $r\alpha$ for arbitrary $r \in R$. If every element of M also belongs to N , then M is said to be divisible by N , written $M \equiv 0 \pmod{N}$; equivalently, M is a multiple and N a divisor. R -modules all of whose elements lie in R are ideals of R .

The intersection, or least common multiple, of any family of modules is again a module, and so is T . Thus the module generated by any system of elements is defined uniquely as the intersection of all modules containing that system. The sum, or greatest common divisor, of a family of modules is the module generated by their union. The product MN of two modules is the module generated by all products $\mu\nu$ with $\mu \in M$ and $\nu \in N$. This product is associative and commutative because the ring multiplication is; distributivity follows from distributivity in T .

Since R itself is a module, saying that R is the multiplier domain is expressed by $RM \equiv 0 \pmod{M}$. Since R contains the identity element, also $M \equiv 0 \pmod{RM}$, hence $M = RM$.

A module is finite if it is generated by finitely many elements. If

$$M = (\mu_1, \dots, \mu_s),$$

then the μ_i form a module basis. Sums and products of finitely many finite modules are again finite; for products this follows from writing every element of M as an R -linear combination of the basis elements and using commutativity of multiplication in T .

An extension ring of R lying in T , that is, a subring of T containing all elements of R , is called an R -order, or simply an order. The intersection of arbitrary orders is again an order, and T is an order; hence the order generated by any set of elements is uniquely defined. Every order is also an R -module. An order is finite if it is finite as a module. Sums and products of finite orders are finite orders. There is also a transitivity law: if A is a finite R -order and B a finite A -order, then B is a finite R -order, since products of an R -basis of A with an A -basis of B give an R -basis of B .

An element $\alpha \in T$ is called integral over R if the order $R[\alpha]$ generated by it is finite. Equivalently, the chain of modules

$$R_1 = (1), \quad R_2 = (1, \alpha), \quad R_3 = (1, \alpha, \alpha^2), \dots$$

terminates. Equivalently again, there exists a non-zero principal module $C = (\gamma)$ such that $CR[\alpha]$ is finite; that is, the chain CR_i terminates. Finally this is identical with the usual equation formulation:

α is integral if it satisfies an equation

$$\alpha^m + r_1\alpha^{m-1} + \cdots + r_m = 0, \quad r_i \in R.$$

These formulations are equivalent. If $R[\alpha]$ is finite, a basis can be chosen among the powers of α , and termination of the chain gives the monic equation. Conversely such an equation expresses every high power of α in terms of lower powers. The principal-module formulation is equivalent because multiplying by a non-zero principal module in a domain does not destroy the eventual equality of the chains.

The ring property of the integral elements and the transitivity of integral dependence rest on the following lemma.

Lemma. If $\mathcal{O}$ is a finite order in T and $\alpha \in \mathcal{O}$, then the order $R[\alpha]$ generated by α is finite. In other words, all elements of a finite order are integral.

Indeed, let $\omega_1, \dots, \omega_s$ be a module basis of $\mathcal{O}$. Since $\alpha\omega_i$ again lies in $\mathcal{O}$, there are relations

$$\alpha\omega_i = \sum_j r_{ij}\omega_j.$$

The determinant of the corresponding system gives a monic equation for α . The use of the determinant is legitimate because T , and hence $\mathcal{O}$, has no zero divisors.

The system $\mathcal{G}$ of all elements of T integral over R is an order. It contains R , because the elements of R are integral with basis e . It is closed under addition and multiplication: if α and β are integral, the order generated by them is the product of two finite orders and is finite, so $\alpha + \beta$ and $\alpha\beta$ are elements of a finite order and hence integral by the lemma.

The transitivity law for integral dependence is this: if $\mathcal{O}$ is an order consisting of elements integral over R , then every element of T integral over $\mathcal{O}$ is already integral over R . For such an element α satisfies a monic equation with coefficients in $\mathcal{O}$; the order generated over R by those finitely many coefficients is finite, and α is integral over that finite order. By the transitivity of finite orders, $R[\alpha]$ is finite.

The ring R is called integrally closed in T if every element of T integral over R already belongs to R . By the second formulation of integrality, an element $\alpha \in T$ belongs to R exactly when there is a non-zero principal module C such that $CR[\alpha]$ is finite.

Dedekind drew two important consequences from integral closedness.

Dedekind consequence I. Suppose R is integrally closed in T , and suppose further that the divisor chain condition for ideals of R holds. If β is not integral over R and $c \neq 0$ is an element of R , then there is an exponent $\rho > 0$ such that in the sequence

$$c, \quad c\beta, \quad c\beta^2, \quad \dots, \quad c\beta^3, \dots$$

the terms $c, \dots, c\beta^{\rho-1}$ are integral and all subsequent terms are not integral.

The point is that if all $c\beta^r$ were integral, then by integral closedness they would all belong to R . The ideals generated by the initial segments would form a divisor chain and would therefore stop. That would imply finiteness of $R[\beta]$ after multiplication by c , hence integrality of β , contrary to assumption. Once a non-integral term occurs, every later term remains non-integral; otherwise a power relation would make the earlier term integral. Thus there is a first non-integral term.

Specializing T to the quotient field of R gives the second consequence.

Dedekind consequence II. Suppose R is integrally closed in its quotient field and satisfies the divisor chain condition for ideals. If β is a non-integral element of the quotient field, then it has a representation

$$\beta = m/n, \quad m, n \in R,$$

such that m^2/n is also not integral.

Indeed, write $\beta = b/c$ with $c \neq 0$. In the sequence $c, c\beta = b, c\beta^2, \dots$ the first two terms are integral, so the exponent ρ of the preceding consequence is greater than one. Put

$$m = c\beta^\rho, \quad n = c\beta^{\rho-1}.$$

Then $m/n = \beta$, while $m^2/n = c\beta^{\rho+1}$ is not integral.

Finally, integral closedness is itself transitive: if $\mathcal{G}$ is the system of elements of T integral over R , then $\mathcal{G}$ is also the system of elements integral over $\mathcal{G}$. Any element integral over $\mathcal{G}$ is integral over R by transitivity, hence belongs to $\mathcal{G}$.

2. Chain conditions in finite module domains

The module theorem used in the application says that chain conditions pass from the coefficient ring to finite modules. A module domain M over a ring R is an abelian group with scalar multiplication by elements of R satisfying the associative and distributive laws. All module definitions from the preceding subsection that do not use multiplication of module elements remain valid.

A module domain is finite if finitely many elements $\varepsilon_1, \dots, \varepsilon_k$ generate it; if R has a unit, its elements are all linear forms

$$r_1\varepsilon_1 + \dots + r_k\varepsilon_k.$$

Module theorem. Let M be a finite module domain over a commutative ring R with identity. If the divisor chain condition, respectively the multiple chain condition, holds for ideals of R , then the corresponding divisor, respectively multiple, chain condition holds for submodules of M .

The proof assigns to each submodule U of M a finite system of ideals $a_1, \dots, a_k$ of R . An element of M is said to have length i if it can be expressed using $\varepsilon_1, \dots, \varepsilon_i$ but not using fewer generators. Let U_i be the elements of U of length at most i , and let a_i be the set of coefficients of ε_i occurring in elements of U_i . If V is a proper divisor or proper multiple of U , then at least one of the ideals

assigned to V is a proper divisor or proper multiple of the corresponding a_i . Thus an infinite chain of submodules would produce an infinite chain among finitely many ideal chains in R , impossible under the corresponding chain condition.

A useful consequence is the following. If in R the multiple chain condition holds modulo every non-zero ideal, then in a finite module domain M the multiple chain condition holds modulo every submodule whose associated ideals $c_1, \dots, c_k$ are all non-zero. The proof is the same, applied modulo those non-zero ideals.

3. Passage to finite extension fields

The preceding module theorem is used to bring algebraic number fields and function fields under the axioms.

Assume that in R all axioms I–V except possibly the multiple chain condition II hold. Let K be the quotient field of R , and let L be a finite extension field of first kind over K . Let $\mathcal{G}$ be the system of elements of L integral over R . Then $\mathcal{G}$ satisfies the same axioms, and every order contained in $\mathcal{G}$ satisfies all of them except possibly integral closedness.

The axioms III and IV hold because the integral elements form a ring without zero divisors and with the same identity. Integral closedness of $\mathcal{G}$ in L follows from the transitivity of integral dependence. It remains to see the divisor chain condition. Since the extension is of first kind, the field L is generated by a single element α integral over R . Its conjugates in the Galois field are integral, and the square D of the product of their differences is a non-zero element of K , hence, by integral closedness of R , an element of R . The usual discriminant argument shows that the integral closure $\mathcal{G}$ is contained in a finite R -module generated by elements of the form α^j/D . Therefore the module theorem gives the divisor chain condition for ideals of $\mathcal{G}$ and for ideals of every order in $\mathcal{G}$.

If all axioms I–V hold in R , then all hold in $\mathcal{G}$, and every order in $\mathcal{G}$ satisfies all but integral closedness. The only remaining point is axiom II. By the consequence of the module theorem it suffices to show that a non-zero ideal of $\mathcal{G}$, regarded as an R -module inside the finite module containing $\mathcal{G}$, has all associated coefficient ideals non-zero. This follows because multiplication by a suitable non-zero element of R carries each basis element of the finite module into the given ideal. The same argument applies to orders of full rank; for lower-rank orders one passes to their quotient fields, intermediate between K and L and again of first kind.

It remains only to verify the axioms for the basic domains: the rational integers, polynomials in one indeterminate, and the functional domain of polynomials in several indeterminates. For the rational integers and for polynomials in one indeterminate over a field, the axioms follow from unique factorization into irreducible elements and from the fact that every ideal is principal. Integral closedness follows because a non-integral quotient can be written with coprime numerator and denominator, so that no power of the numerator is divisible by the denominator.

For the ring of polynomials in several indeterminates, axioms III, IV, and V again follow from unique

factorization of elements. The divisor chain condition is Hilbert's basis theorem. The multiple chain condition need not hold. Passing to the functional domain restores it. One adjoins a new indeterminate u and considers rational functions whose denominators are primitive polynomials in u . The elements with primitive numerator and denominator become units, and every element of the functional domain has a unique expression as an element of the original polynomial ring multiplied by a unit. The factorization theory of the original ring therefore transfers to the functional domain. In that domain every ideal is principal: starting with an element of an ideal and another not divisible by it, the greatest common divisor also lies in the ideal and is a proper divisor; after finitely many repetitions one obtains a basis element. Hence unique factorization of ideals and the axioms I and II follow.

4. Isomorphism theorems and direct sums

In the following only the ring or module structure is assumed; no chain axiom is required.

Let M and $\bar{M}$ be module domains over R . We say that M is homomorphic to $\bar{M}$ if to every element of M there corresponds one and only one element of $\bar{M}$, so that all of $\bar{M}$ is reached, and such that difference and multiplication by the same element of R correspond. Thus from

$$\beta \sim \bar{\beta}, \quad \gamma \sim \bar{\gamma}$$

one obtains

$$\beta - \gamma \sim \bar{\beta} - \bar{\gamma}, \quad r\beta \sim r\bar{\beta}.$$

Every R -module B in M then corresponds to an R -module $\bar{B}$ in $\bar{M}$. Conversely, if $\bar{C}$ is a module in $\bar{M}$, the totality C^* of elements of M to which elements of $\bar{C}$ correspond is an R -module in M uniquely determined by $\bar{C}$; it is the module associated with $\bar{C}$. If A is the module corresponding to the zero element of $\bar{M}$, then C^* is a divisor of M and decomposes into congruence classes modulo A , these classes corresponding one-to-one to the elements of $\bar{C}$. Starting from a module B in M , passing to $\bar{B}$ in $\bar{M}$, and coming back gives

$$B^* = (B, A),$$

the greatest common divisor of B and A . If B divides M , then $B^* = B$.

If the correspondence is reversible, the module domains are isomorphic. If A is any submodule of M , the residue-class module M/A is obtained by taking congruence modulo A as the equality relation. Equivalently, one replaces equal elements modulo A by one residue class. Every homomorphism is obtained in this way: if M maps homomorphically onto $\bar{M}$ and A is the kernel, then $\bar{M}$ is isomorphic to M/A .

The two isomorphism theorems follow. First, if $\bar{M} = M/A$ and C divides A , then

$$\bar{M}/C \simeq M/C.$$

Indeed, equality modulo A is already equality modulo C after the indicated passage to residue classes. Secondly, for two submodules B and A of M one has

$$(B, A)/A \simeq B/[B, A],$$

where (B, A) denotes the greatest common divisor and $[B, A]$ the least common multiple. This is the module form of the familiar second isomorphism theorem.

The same considerations apply to commutative rings when modules are replaced by ideals. If R maps homomorphically onto $\bar{R}$, difference and product correspond. If $\mathfrak{a}$ is an ideal, the residue-class ring $R/\mathfrak{a}$ is formed by using congruence modulo $\mathfrak{a}$ as equality; products of residue classes are then well-defined. Every ring homomorphism is obtained in this way, and the isomorphism theorems become

$$(R/\mathfrak{a})/(\mathfrak{c}/\mathfrak{a}) \simeq R/\mathfrak{c} \quad (\mathfrak{c} \mid \mathfrak{a}),$$

and

$$(\mathfrak{b}, \mathfrak{a})/\mathfrak{a} \simeq \mathfrak{b}/[\mathfrak{b}, \mathfrak{a}].$$

Assume now that the commutative ring R has a multiplicative identity. Two ideals $\mathfrak{a}, \mathfrak{b}$ are relatively prime if their greatest common divisor is the unit ideal,

$$(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}.$$

The basic rules are most simply stated without writing identity elements explicitly. If $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$, then

$$(\mathfrak{a}, \mathfrak{b}\mathfrak{c}) = (\mathfrak{a}, \mathfrak{c}).$$

Therefore from $\mathfrak{c}\mathfrak{b} \equiv 0 \pmod{\mathfrak{a}}$ and $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{o}$ it follows that $\mathfrak{c} \equiv 0 \pmod{\mathfrak{a}}$. If every $\mathfrak{a}_i$ is relatively prime to every $\mathfrak{b}_j$, then the product of the $\mathfrak{a}_i$ is relatively prime to the product of the $\mathfrak{b}_j$. In particular, if $\mathfrak{b}_1, \dots, \mathfrak{b}_s$ are pairwise relatively prime, their least common multiple is their product.

The Chinese-remainder decomposition is then immediate. If $\mathfrak{b}_1, \dots, \mathfrak{b}_s$ are pairwise relatively prime and

$$\mathfrak{v} = [\mathfrak{b}_1, \dots, \mathfrak{b}_s] = \mathfrak{b}_1 \cdots \mathfrak{b}_s,$$

then the residue-class ring modulo $\mathfrak{v}$ is the direct sum of the residue-class rings modulo the $\mathfrak{b}_i$:

$$R/\mathfrak{v} \simeq R/\mathfrak{b}_1 \oplus \cdots \oplus R/\mathfrak{b}_s.$$

The idempotents giving the direct summands are obtained by choosing, for each i , an element congruent to 1 modulo $\mathfrak{b}_i$ and congruent to 0 modulo the other $\mathfrak{b}_j$.

Conversely, a decomposition of a ring or module into a direct sum is expressed ideal-theoretically by such idempotent decompositions and by pairwise relatively prime annihilator ideals. This is the form in which Noether uses direct sums later: it is not an additional structure, but the ideal-theoretic

expression of pairwise relatively prime components.

5. Primary ideals without chain assumptions

Noether distinguishes weak and strong primary ideals in a general ring before imposing chain conditions. An ideal $\mathfrak{q}$ is weakly primary if, whenever $ab \equiv 0 \pmod{\mathfrak{q}}$ and a is not divisible by $\mathfrak{q}$, every sufficiently high power of b is divisible by $\mathfrak{q}$. Equivalently, the zero divisors of the residue-class ring $R/\mathfrak{q}$ are precisely those residue classes all of whose sufficiently high powers vanish. The associated prime ideal $\mathfrak{p}$ consists of all elements whose powers are divisible by $\mathfrak{q}$.

The strong version is obtained by replacing elements by ideals. Thus $\mathfrak{q}$ is strongly primary if from

$$\mathfrak{a}\mathfrak{b} \equiv 0 \pmod{\mathfrak{q}}$$

and from the non-divisibility of $\mathfrak{a}$ by $\mathfrak{q}$, it follows that some power of $\mathfrak{b}$ is divisible by $\mathfrak{q}$. In rings with the divisor chain condition the distinction disappears; without that hypothesis it is real. Noether records an example, due to R. Hölzer, showing that a weak primary ideal need not be strong.

The least common multiple of finitely many weak, respectively strong, primary ideals with the same associated prime ideal is again weak, respectively strong, primary with that same associated prime ideal. But a shortest representation by finitely many primary ideals belonging to different prime ideals is not primary. Indeed, if

$$\mathfrak{m} = [\mathfrak{q}_1, \dots, \mathfrak{q}_r]$$

is a shortest representation with distinct associated primes $\mathfrak{p}_i$, one can choose elements, or ideals in the strong case, which vanish modulo all but one component and not modulo the remaining prime. Multiplying them gives a product divisible by $\mathfrak{m}$ while no suitable power condition required for primaryness can hold. This proves non-primaryness of such a mixed component.

The quotient of modules is now introduced. Let T be an extension ring of R , and let $M, B, C, \dots$ be R -modules in T . The quotient

$$C = M : B$$

is the smallest module satisfying

$$CB \equiv 0 \pmod{M},$$

in the sense that if $DB \equiv 0 \pmod{M}$, then $D \equiv 0 \pmod{C}$. Equivalently, C is the greatest common divisor of all modules D with $DB \equiv 0 \pmod{M}$. The zero module always gives existence. From the definition one obtains the formal rules: if M is a multiple of N , then $M : B$ is a multiple of $N : B$; if B is a multiple of C , then $M : B$ is a divisor of $M : C$; and

$$M : BC = (M : B) : C = (M : C) : B.$$

If the extension ring is the original ring, the module quotient becomes the ideal quotient $\mathfrak{a} : \mathfrak{b}$, with

the same rules. Since $\mathfrak{a}\mathfrak{b} \equiv 0 \pmod{\mathfrak{a}}$, one also has

$$\mathfrak{a} \equiv 0 \pmod{\mathfrak{a} : \mathfrak{b}}.$$

An ideal $\mathfrak{b}$ is called prime to $\mathfrak{a}$ if

$$\mathfrak{a} : \mathfrak{b} = \mathfrak{a}.$$

This says exactly that $d\mathfrak{b} \equiv 0 \pmod{\mathfrak{a}}$ always implies $d \equiv 0 \pmod{\mathfrak{a}}$. From the quotient rules it follows that if $\mathfrak{b}$ and $\mathfrak{c}$ are prime to $\mathfrak{a}$, then both their product and their least common multiple are prime to $\mathfrak{a}$. If $\mathfrak{b}$ is prime to $\mathfrak{a}$ and $\mathfrak{a}$ is prime to $\mathfrak{b}$, the ideals are mutually prime. Relatively prime ideals in the sense of greatest common divisor one are mutually prime, but the quotient formulation is more flexible and is the one needed for the proofs under chain conditions.

6. Ideal theory under the divisor chain condition

Assume now that the commutative ring R satisfies axiom I, the divisor chain condition. Fix once and for all a well-ordering of all elements of R . This also gives a well-ordering of the ideals. Indeed, the divisor chain condition implies that every ideal has a finite basis; ordering finite subsets quasi-lexicographically and assigning to each ideal its first possible basis well-orders the set of ideals.

Theorem I. Under the divisor chain condition every ideal of R can be represented as a least common multiple of finitely many irreducible ideals, i.e. ideals not representable as a least common multiple of two proper divisors.

If the theorem failed for an ideal $\mathfrak{m}$, then $\mathfrak{m}$ would be reducible; writing $\mathfrak{m} = [\mathfrak{a}, \mathfrak{b}]$, at least one of $\mathfrak{a}$ or $\mathfrak{b}$ would again fail the theorem. Choose the first such proper divisor in the well-ordering. Repeating the construction gives a definite infinite chain of proper divisors, contradicting the divisor chain condition. This is the only part of the argument where the well-ordering is used; the statement is set-theoretic in nature and applies equally to module domains with the corresponding chain condition.

With the divisor chain condition, primary ideals in the weak and strong sense coincide, and one may simply speak of primary ideals. The relation between primary and irreducible is the key.

Theorem II. Under the divisor chain condition every irreducible ideal is primary. Equivalently, every non-primary ideal is reducible.

If $\mathfrak{m}$ is not primary, then there are elements, or equivalently ideals through the quotient formalism, giving a product divisible by $\mathfrak{m}$ while neither factor satisfies the required power-divisibility condition. The ideal quotient then furnishes two proper divisors of $\mathfrak{m}$ whose least common multiple is $\mathfrak{m}$ itself. Thus $\mathfrak{m}$ is reducible. Combining this with Theorem I gives the representation of every ideal as a least common multiple of finitely many primary ideals.

Continuing the same proof, once every irreducible ideal is known to be primary, the representation

theorem takes the following explicit form. Every ideal $\mathfrak{a}$ admits a shortest representation

$$\mathfrak{a} = [\mathfrak{q}_1, \dots, \mathfrak{q}_s]$$

by primary ideals. The associated prime ideals are obtained by taking, for each $\mathfrak{q}_i$, the totality of elements whose powers are divisible by $\mathfrak{q}_i$. The primary components may still fail to be unique at this stage. What is fixed by the divisor chain condition alone is the existence of a finite primary representation and the possibility of choosing it shortest.

Noether's reason for introducing the ideal quotient now becomes visible. If $\mathfrak{b}$ is prime to $\mathfrak{a}$, the equality $\mathfrak{a} : \mathfrak{b} = \mathfrak{a}$ is an exact algebraic substitute for saying that $\mathfrak{b}$ does not meet the component defined by $\mathfrak{a}$. Multiplying by ideals prime to a component cannot introduce new divisibility modulo that component. Thus the quotient calculus supplies the separation of components without choosing elements or bases.

The next step, which belongs to the multiple chain condition, is the uniqueness of isolated components. Passing to the residue-class ring modulo a non-zero ideal, the multiple chain condition prevents a prime ideal from having proper divisors other than the unit ideal. Consequently the primary component corresponding to an isolated associated prime is forced: it is recovered as the quotient of the given ideal by all ideals prime to that associated prime. This is the abstract version of separating one primary component by multiplying away all the others.

When the ring has an identity, the unit ideal no longer appears as an associated prime for non-zero ideals. Hence the ambiguity just described disappears for non-zero ideals once both chain conditions hold. The primary components become unique; they are pairwise relatively prime, and their least common multiple is their product. Thus the theory has reached the form familiar from orders in number fields: an ideal is represented as a product of one-type components, and after integral closedness is added these one-type components become powers of prime ideals.

This order of proof is important. The primary decomposition, its uniqueness, and the passage from primary ideals to prime powers are not bundled together. They are separated according to the axioms that force them: divisor chains give existence, multiple chains give uniqueness of the components, the identity and absence of zero divisors remove exceptional unit phenomena and create a quotient field, and integral closedness turns primary components into prime powers. The classical Dedekind ideal theory is therefore not a single indivisible phenomenon but the conjunction of these four structural facts.

For the continuation of the paper this separation of the axioms is the controlling device. In the next sections Noether proves the converse direction: from the prime-power factorization of ideals one recovers the chain conditions and integral closedness. The proof is again conducted with ideal quotients. Fractional ideals are introduced as quotients in the quotient field, their classes form a group precisely when every non-zero ideal factors into prime powers, and Dedekind's criterion for integral closedness is recovered from the impossibility of writing a genuinely fractional element with all powers of the numerator divisible by the denominator. Thus the abstract axioms and the

classical factorization theorem determine one another.

This is the point at which the paper passes from mere primary decomposition to Dedekind ideal theory. The previous sections account for the decomposition of ideals into components; the subsequent ones show why, in integrally closed domains satisfying the double chain condition, those components have the special and much stronger form of powers of prime ideals. The proof is deliberately intrinsic: it does not rely on the existence of bases for ideals in a number field, but only on divisibility, quotients, chain conditions, and integrality.

The distinction is also why the paper can include both algebraic number fields and algebraic function fields without changing the formal proof. In the number-field case the integral closure of the rational integers in a finite extension field is the standard order of all algebraic integers. In the function-field case the functional domain plays the same formal role after the passage that removes the lower-dimensional ideals. Once the axioms are verified for the base domain and shown to survive finite extensions of first kind, the subsequent ideal theory proceeds without referring to the concrete origin of the ring.

The practical mathematical point is that Noether is not merely restating Dedekind's theory abstractly. She is isolating exactly which parts of the classical proof require which finiteness or integrality assumptions. This separation lets later commutative algebra treat primary decomposition, isolated components, ideal quotients, integral closure, and unique factorization of ideals as separate structural layers rather than as one monolithic theorem inherited from algebraic number theory.